Missed ACS Codes Review

Nine knowledge codes, one lesson

Cessna 172S NAV III (G1000)

FAA-S-ACS-25

[Name], CFI Applicant



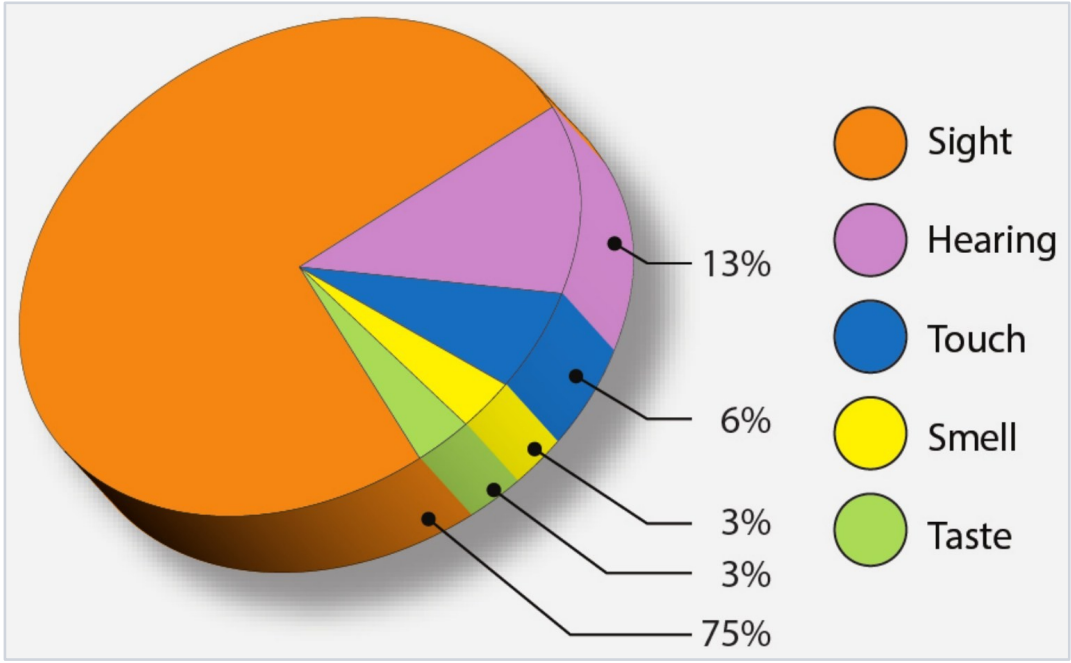
Source: PHAK Fig 16-29

Objective

Review

- **Explain each missed code**
- **Use FAA wording**
- **Tie it to the C172S**
- **Teach it back clearly**

State the objective: explain each of the nine knowledge elements using FAA wording, apply it to the C172S, and teach it back.



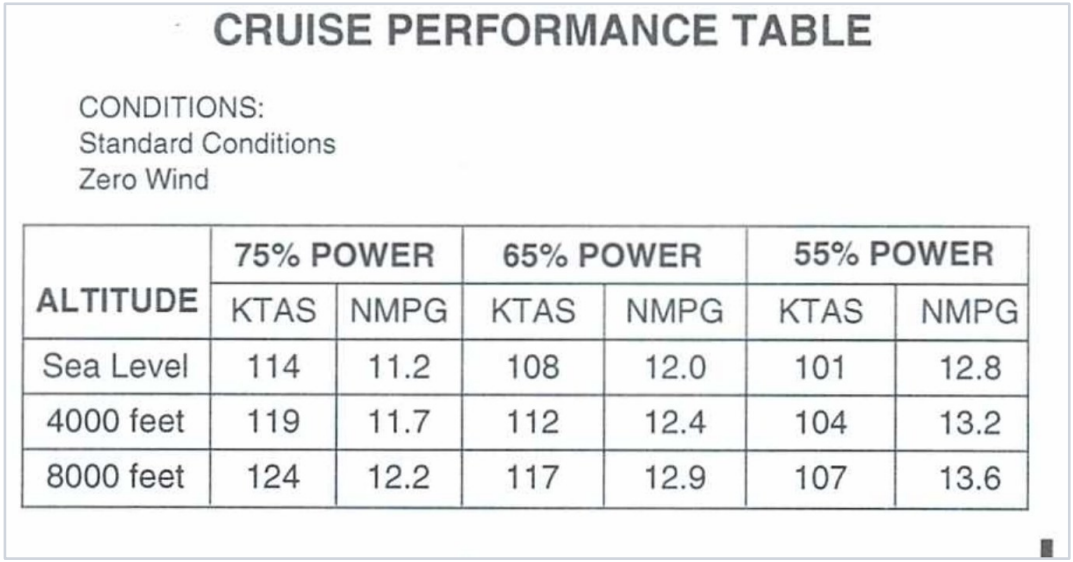
Source: ACS FI.I.B, ACS AI.II, ACS AI.III.B, ACS AI.XII.D

The Nine Codes

Review

Code	Topic
FI.I.B.K3	Perceptions and insight
AI.II.D.K2	Stability and control
AI.II.E.K1c	Powerplant and propeller
AI.II.F.K3	Weight and balance terms
AI.II.H.K1	Ground-based navigation
AI.II.I.K3	Plotting a course
AI.II.I.K4	Power setting selection
AI.III.B.K1b	Inspections and logbooks
AI.XII.D.K1	ELT

Walk through the list. Each code gets two to five slides: the FAA definition, plain words, and a C172S example.



CRUISE PERFORMANCE TABLE

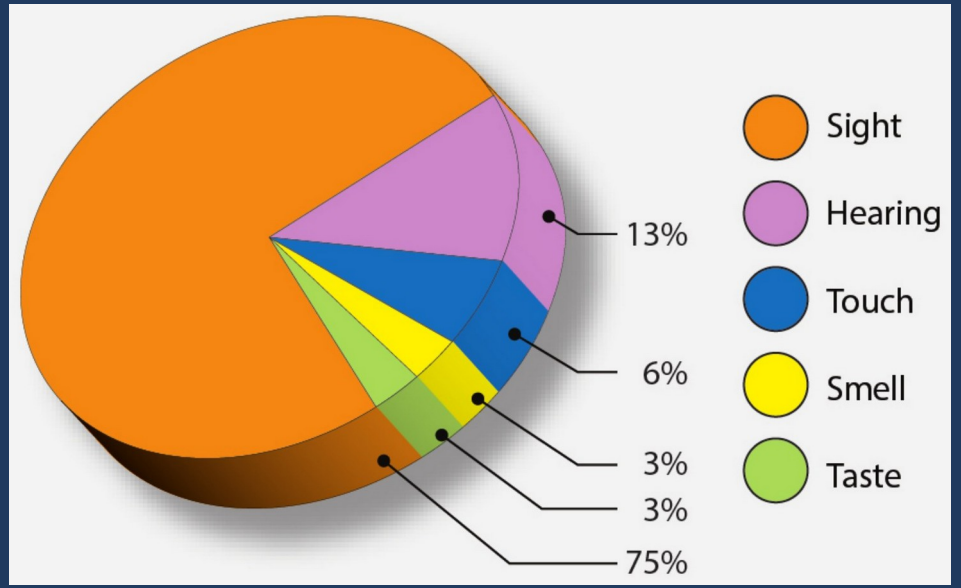
CONDITIONS:
Standard Conditions
Zero Wind

ALTITUDE	75% POWER		65% POWER		55% POWER	
	KTAS	NMPG	KTAS	NMPG	KTAS	NMPG
Sea Level	114	11.2	108	12.0	101	12.8
4000 feet	119	11.7	112	12.4	104	13.2
8000 feet	124	12.2	117	12.9	107	13.6

Source: FAA-S-ACS-25

FI.I.B.K3

Perceptions and Insight



What you will learn

- What perceptions are
- Five factors that affect perception
- What insight is
- How you build insight

Why it matters: every lesson you teach goes in through your student's senses. Knowing what blocks perception makes you a better instructor.

Source: AIH 3-7 to 3-9

Definition: Perceptions

FI.I.B.K3

PERCEPTIONS

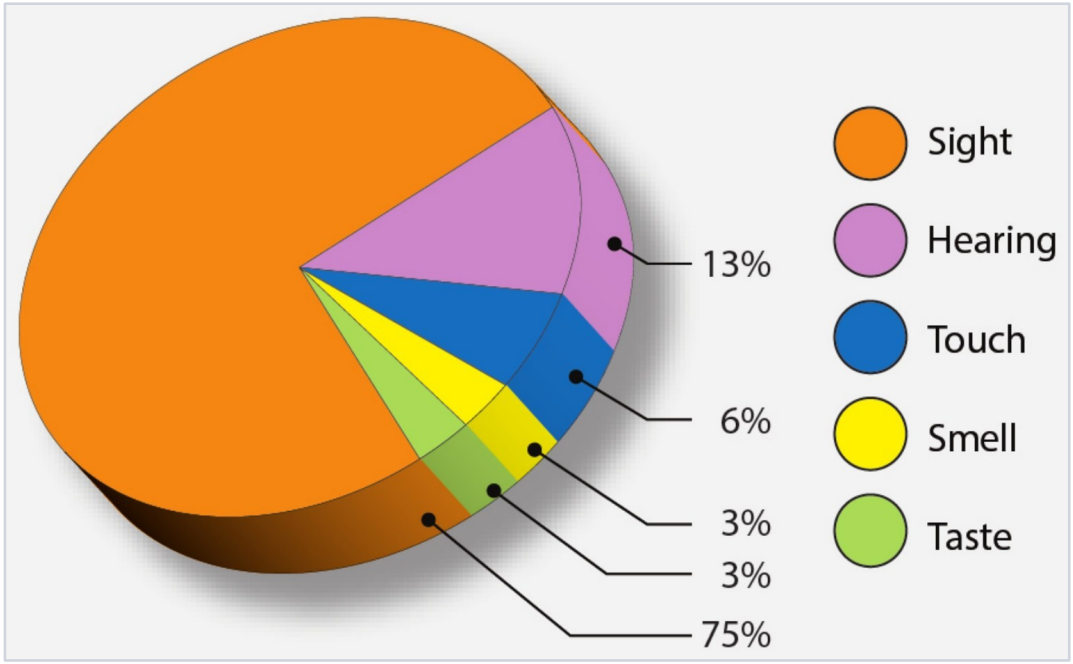
"Initially, all learning comes from perceptions, which are directed to the brain by one or more of the five senses: sight, hearing, touch, smell, and taste."

MORE THAN ONE SENSE

"learning occurs most rapidly when information is received through more than one sense."

IN PLAIN WORDS

Everything a student learns comes in through the senses. Show it, say it, and let them touch it, and it sticks faster.



KNOW THIS FOR THE CHECKRIDE

Sight is about 75 percent of learning. Sight and hearing together are about 88 percent (AIH Fig 3-5).

Source: AIH 3-7, AIH Fig 3-5

Factors That Affect Perception

FI.I.B.K3

Factor	Plain words
Physical organism	Senses must work
Goals and values	What they care about
Self-concept	Confident or not
Time and opportunity	Time and right order
Element of threat	Fear narrows focus

AIH 3-7 lists five factors: physical organism, goals and values, self-concept, time and opportunity, and element of threat. The study guide memory aid GSTEP covers the same five in a different order (GSTEP is not an AIH term).

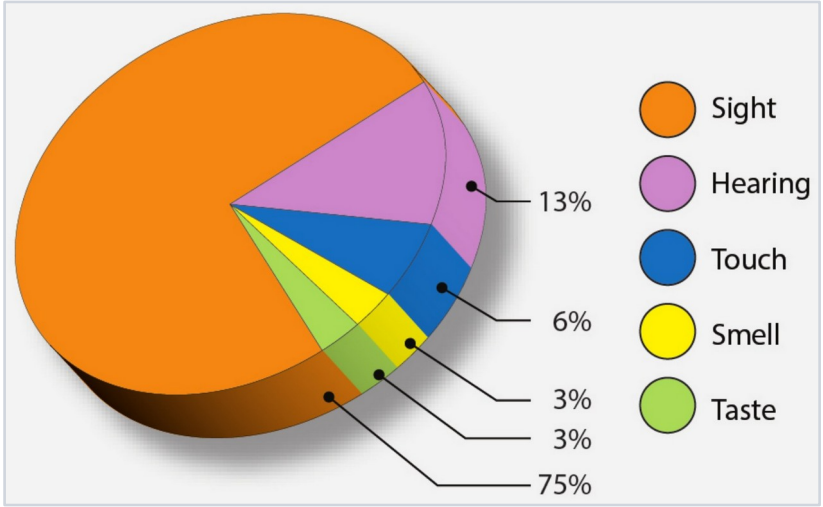
Physical organism: the senses a pilot needs to see, hear, feel, and respond (AIH 3-8).

Goals and values: perceptions depend on what the learner values (AIH 3-8).

Self-concept: a positive self-image keeps the learner open to new experiences. A negative one creates barriers (AIH 3-8).

Time and opportunity: it takes time and the right order of experiences to perceive (AIH 3-8).

Element of threat: fear narrows the perceptual field, so threats do not help learning (AIH 3-8).



MEMORY AID

GSTEP (from your study guide): Goals and values, Self-concept, Time and opportunity, Element of threat, Physical organism.

EXAMPLE

AIH example: during the first steep turns, a learner may stare at the altimeter and ignore outside references. That is the element of threat narrowing perception (AIH 3-8).

Source: AIH 3-7, AIH 3-8

Definition: Insight

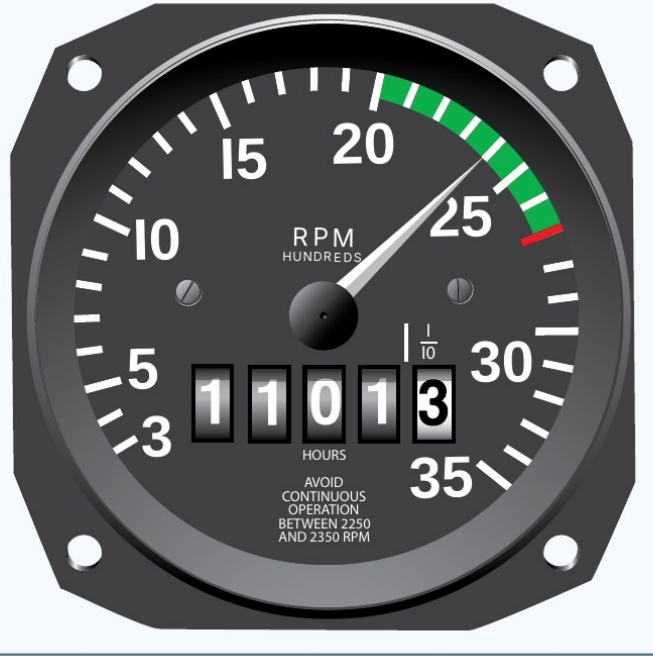
FI.I.B.K3

INSIGHT

“Insight involves the grouping of perceptions into meaningful wholes. Creating insight is one of the instructor's major responsibilities.”

IN PLAIN WORDS

Insight is the moment the pieces click. The student stops seeing throttle, RPM, and pitch as separate facts and sees how they work together.



EXAMPLE

AIH example made for the C172S: with a fixed-pitch propeller, rpm goes up when the throttle is opened and down when it is closed. Rpm also changes when pitch attitude changes with no power change. Engine speed, power, airspeed, and attitude are all related (AIH 3-9).

Source: AIH 3-9, PHAK Fig 7-8

How Instructors Build Insight

FI.I.B.K3

- **Point out relationships**
- **Keep the learner receptive**
- **Nonthreatening environment**
- **Protect a good self-concept**

Key steps from AIH 3-9: point out relationships as they occur, provide a secure and nonthreatening environment, and help the learner keep a favorable self-concept.

Insight almost always happens eventually, but instruction speeds it up by teaching how perceptions relate as they occur (AIH 3-9).



Source: AIH 3-9

Quick Check

FI.I.B.K3

1 **Which sense accounts for most learning?**

Answer: Sight, about 75 percent (AIH Fig 3-5).

2 **Your student stares at the altimeter during their first steep turns. Which factor is at work?**

Answer: Element of threat. Fear narrows perception (AIH 3-8).

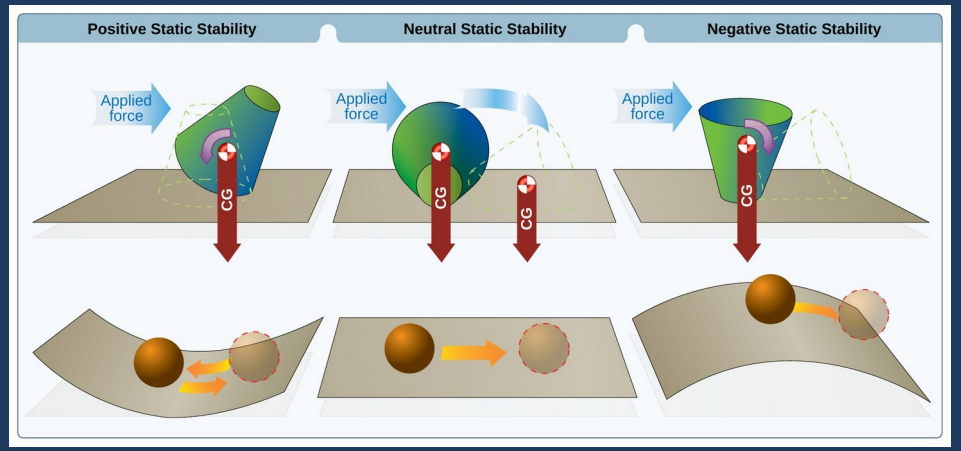
3 **What is insight?**

Answer: Grouping perceptions into meaningful wholes (AIH 3-9).

Source: AIH 3-7 to 3-9, AIH Fig 3-5

AI.II.D.K2

Stability, Maneuverability, and Controllability



What you will learn

- Static vs. dynamic stability
- Positive, neutral, negative
- Maneuverability and controllability
- How CG changes all three

Why it matters: this is why the C172S settles back down after a bump, and why loading it wrong makes it harder to fly.

Source: PHAK 5-14, PHAK 5-15

Definition: Stability

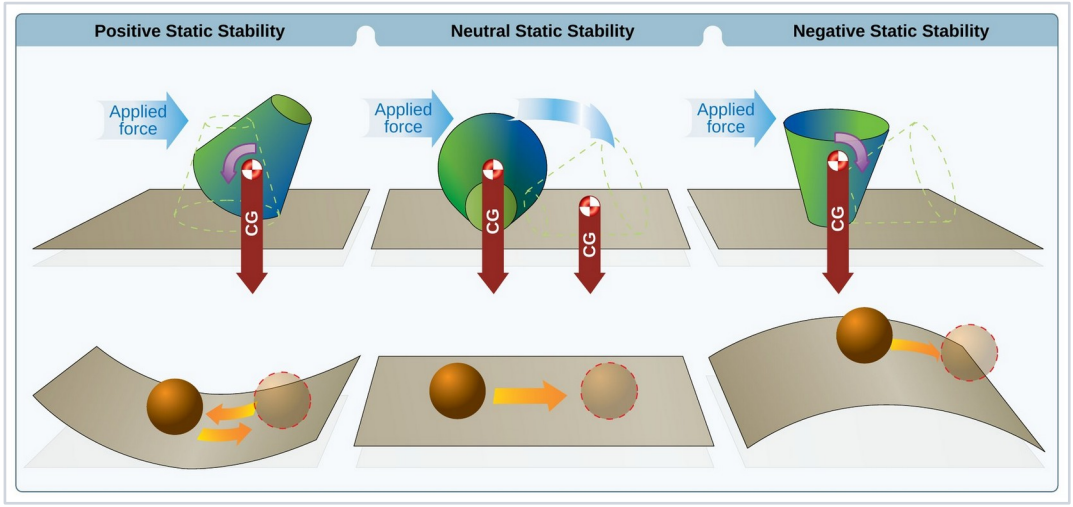
AI.II.D.K2

STABILITY

“Stability is the inherent quality of an aircraft to correct for conditions that may disturb its equilibrium and to return to or to continue on the original flight path. It is primarily an aircraft design characteristic.”

IN PLAIN WORDS

A stable airplane wants to go back to what it was doing after a bump. The designer builds this in.



Source: PHAK 5-14, PHAK Fig 5-21

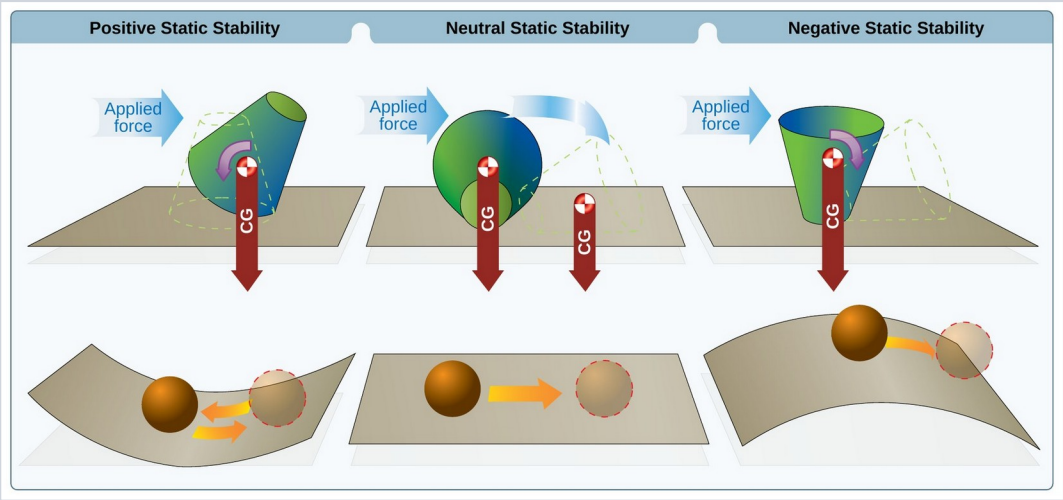
Static Stability

AI.II.D.K2

- **Initial tendency after disturbance**
- **Positive:** returns to equilibrium
- **Neutral:** stays in new condition
- **Negative:** continues away

Static stability is the initial tendency, or direction of movement, back to equilibrium (PHAK 5-14).

Positive static stability: the initial tendency to return to the original state of equilibrium. Neutral: the tendency to remain in a new condition. Negative: the tendency to continue away from the original state (PHAK 5-14).



MEMORY AID

Ball in a bowl rolls back: positive. Ball on a table stays put: neutral. Ball on a hill rolls away: negative.

THINK OF IT LIKE THIS

the PHAK figure shows a ball in a bowl (positive), on a flat surface (neutral), and on top of a hill (negative).

Source: PHAK 5-14, PHAK Fig 5-21

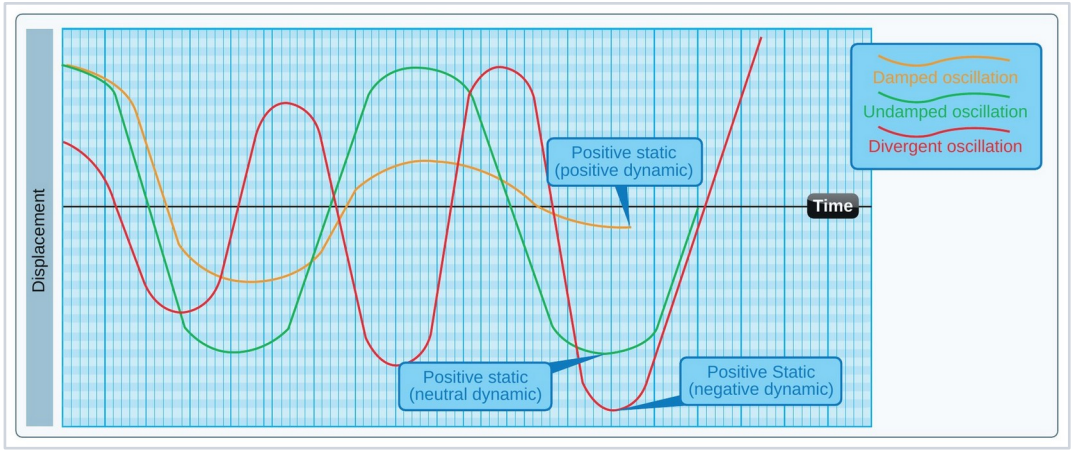
Dynamic Stability

AI.II.D.K2

- **Response over time**
- **Positive:** motion gets smaller
- **Neutral:** motion stays the same
- **Negative:** motion grows

Dynamic stability refers to the aircraft response over time when disturbed (PHAK 5-14).

Positive: the motion decreases in amplitude and returns toward equilibrium. Neutral: the motion neither decreases nor increases. Negative: the motion increases and becomes more divergent (PHAK 5-15).



KNOW THIS FOR THE CHECKRIDE

Static stability is the first reaction. Dynamic stability is what happens over time.

THINK OF IT LIKE THIS

a worn automobile shock absorber shows neutral dynamic stability. The car keeps bouncing the same amount (PHAK 5-15).

Source: PHAK 5-14, PHAK 5-15, PHAK Fig 5-22

Definitions: Maneuverability and Controllability

AI.II.D.K2

MANEUVERABILITY

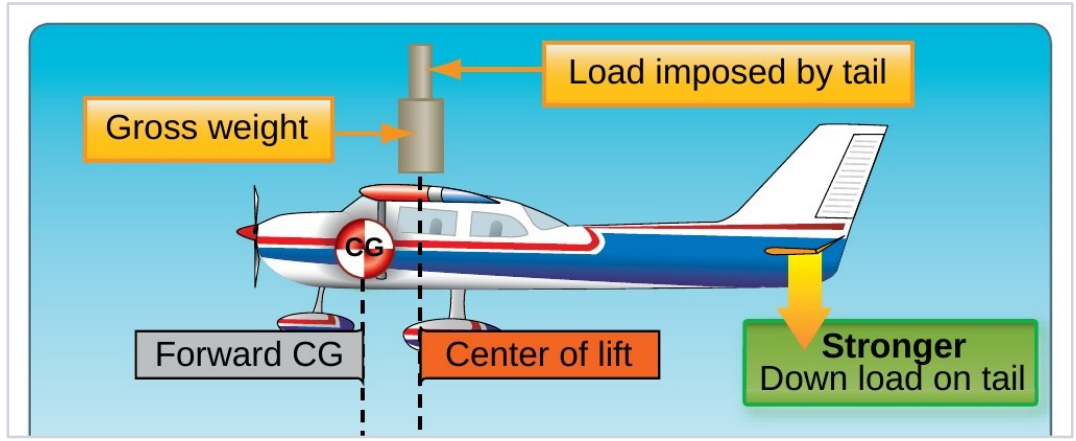
"the quality of an aircraft that permits it to be maneuvered easily and to withstand the stresses imposed by maneuvers. It is governed by the aircraft's weight, inertia, size and location of flight controls, structural strength, and powerplant."

CONTROLLABILITY

"the capability of an aircraft to respond to the pilot's control, especially with regard to flight path and attitude."

IN PLAIN WORDS

Maneuverability: how easily the airplane can be maneuvered and handle the stress. Controllability: how well it does what the pilot asks.



Source: PHAK 5-15, PHAK Fig 5-63

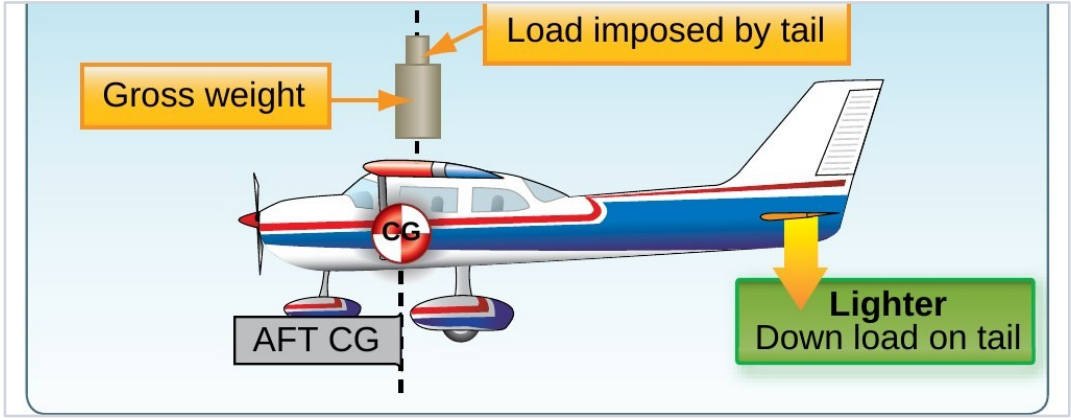
Putting the Three Together

AI.II.D.K2

- **Stability:** wants to return
- **Maneuverability:** maneuvers easily
- **Controllability:** answers the pilot
- **CG location changes all three**

A trainer like the C172S is quick to respond to control inputs, while a transport feels heavy and responds slowly. Designers balance these qualities for the airplane's purpose (PHAK 5-14).

CG location ties it together: an aircraft becomes less controllable at slow speeds and less stable as the CG moves aft (PHAK 5-43, PHAK 5-44).



Source: PHAK 5-15, PHAK 5-43, PHAK 5-44

Quick Check

AI.II.D.K2

1 **A ball resting on a flat table shows which static stability?**

Answer: Neutral. It stays in its new position (PHAK 5-14).

2 **The airplane oscillates and each swing gets bigger. What is that?**

Answer: Negative dynamic stability (PHAK 5-15).

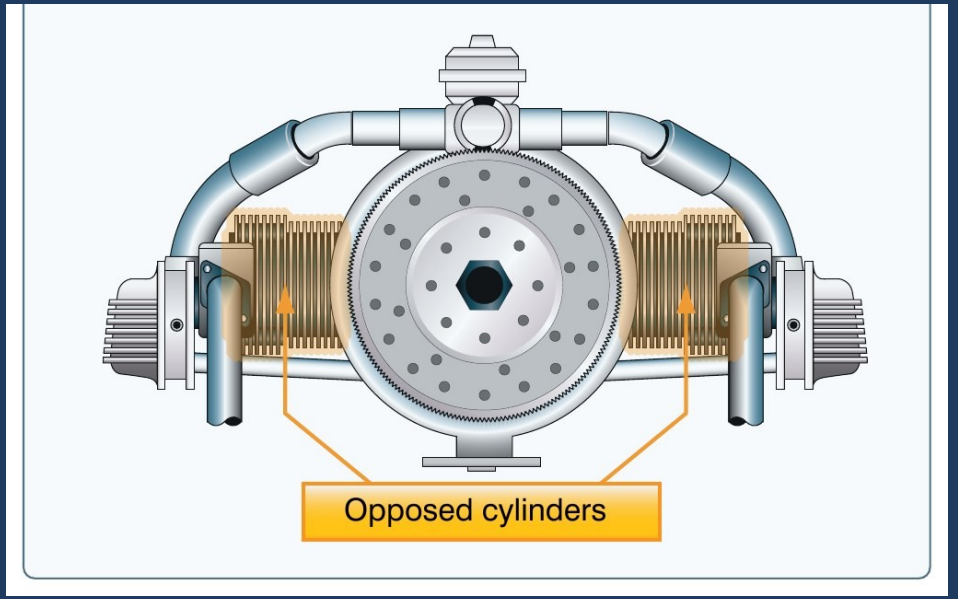
3 **The CG moves aft. More stable or less stable?**

Answer: Less stable (PHAK 5-44).

Source: PHAK 5-14, PHAK 5-15, PHAK 5-44

AI.II.E.K1c

Powerplant and Propeller



What you will learn

- The C172S engine
- Throttle and mixture
- How a propeller works
- Fixed pitch and the tachometer

Why it matters: you have to explain the engine and prop of the airplane you fly, and the C172S is simpler than most study guides describe.

Source: POH 7-28, POH 7-36, PHAK 7-1 to 7-5

C172S Engine

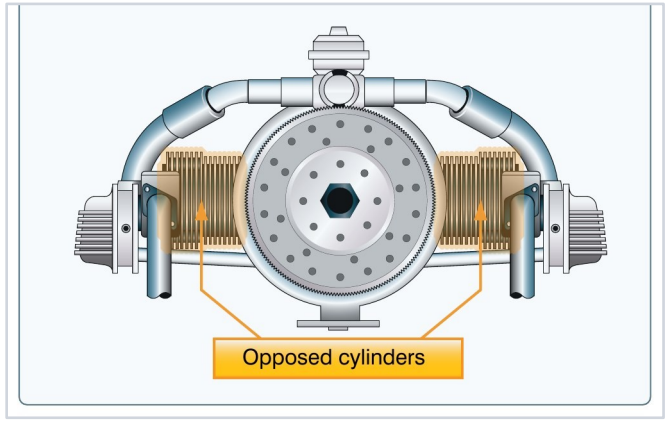
AI.II.E.K1c

- **Lycoming IO-360-L2A**
- **180 hp at 2,700 RPM**
- **Four cylinder, horizontally opposed**
- **Fuel injected, air cooled**

An aircraft engine, or powerplant, produces thrust to propel an aircraft. Reciprocating engines work with a propeller to produce thrust (PHAK 7-1).

C172S: horizontally opposed, four cylinder, overhead valve, air-cooled, fuel-injected engine with a wet sump lubrication system. Lycoming IO-360-L2A rated at 180 horsepower at 2,700 RPM (POH 7-28, TCDS 3A12 p30).

Horizontally opposed engines have high power-to-weight ratios and a small frontal area (PHAK 7-2).



ENGINE

The airplane is powered by a horizontally opposed, four cylinder, overhead valve, air-cooled, fuel-injected engine with a wet sump lubrication system. The engine is a Lycoming Model IO-360-L2A and is rated at 180 horsepower at 2700 RPM. Major accessories include a starter and belt driven alternator mounted on the front of the engine, and dual magnetos, dual vacuum pumps, and a full flow oil filter mounted on the rear of the engine accessory case.

KNOW THIS FOR THE CHECKRIDE

Lycoming IO-360-L2A, 180 hp at 2,700 RPM, four cylinder, horizontally opposed, fuel injected (POH 7-28).

Source: POH 7-28, TCDS 3A12 p30, PHAK 7-2

Engine Controls

AI.II.E.K1c

- **Throttle:** fuel-air flow, RPM
- **Mixture:** red knob
- **Mixture full aft:** idle cutoff
- **No propeller control**

Throttle: open full forward, closed full aft. In a fixed-pitch airplane, rpm is regulated by the throttle, which controls the fuel-air flow to the engine (POH 7-28, PHAK 7-5).

Mixture: red knob with a lock button. Rich is full forward, idle cutoff is full aft (POH 7-28).

Study guide fix: the guide says the throttle regulates fuel flow and lists a propeller control. The PHAK says fuel-air flow, and the C172S has a fixed-pitch propeller with no propeller control.

ENGINE

The airplane is powered by a horizontally opposed, four cylinder, overhead valve, air-cooled, fuel-injected engine with a wet sump lubrication system. The engine is a Lycoming Model IO-360-L2A and is rated at 180 horsepower at 2700 RPM. Major accessories include a starter and belt driven alternator mounted on the front of the engine, and dual magnetos, dual vacuum pumps, and a full flow oil filter mounted on the rear of the engine accessory case.

KNOW THIS FOR THE CHECKRIDE

The throttle controls fuel-air flow and RPM. The C172S has no propeller control.

Source: POH 7-28, PHAK 7-5

Definitions: Propeller and Fixed Pitch

AI.II.E.K1c

PROPELLER

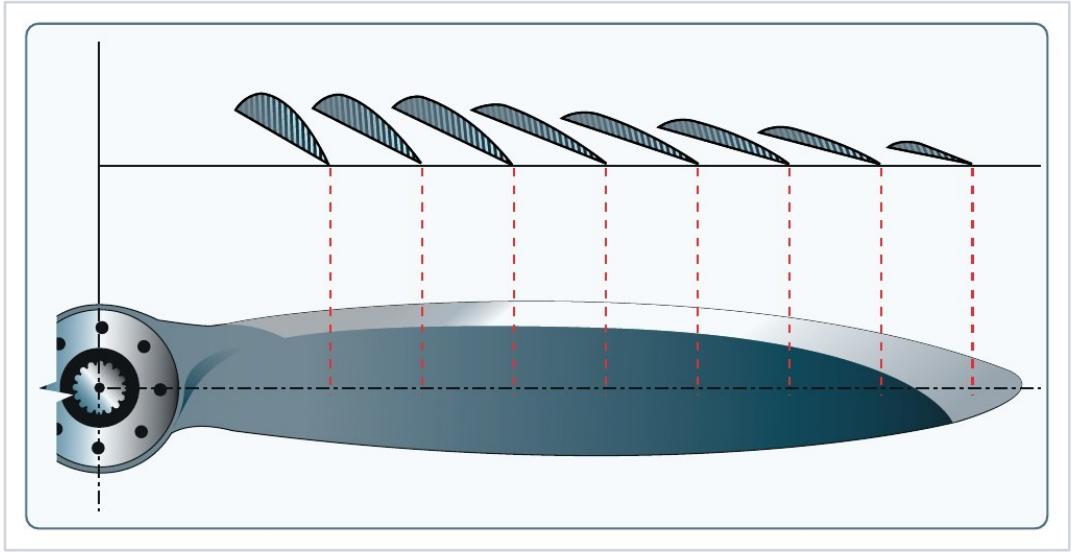
“The propeller is a rotating airfoil, subject to induced drag, stalls, and other aerodynamic principles that apply to any airfoil.”

FIXED-PITCH PROPELLER

“A propeller with fixed blade angles is a fixed-pitch propeller. The pitch of this propeller is set by the manufacturer and cannot be changed.”

IN PLAIN WORDS

The propeller is a spinning wing that pulls the airplane forward. On the C172S the blade angle is set at the factory and never changes.



Source: PHAK 7-4, PHAK 7-5, PHAK Fig 7-6

C172S Propeller

AI.II.E.K1c

- **Two blade, fixed pitch**
- **76 inch forged aluminum**
- **Best at one speed, rpm**
- **Tachometer shows power**

C172S: two bladed, fixed pitch, one-piece forged aluminum alloy propeller, 76 inches in diameter (POH 7-36).

A fixed-pitch propeller is most efficient only at one combination of airspeed and rpm, so it is ideal for neither cruise nor climb. It is used when low weight, simplicity, and low cost are needed (PHAK 7-5).

Propeller efficiency is the ratio of thrust horsepower to brake horsepower (PHAK 5-29). In plain words: the engine makes power, and the propeller turns part of it into thrust.

PROPELLER

The airplane is equipped with a two bladed, fixed pitch, one-piece forged aluminum alloy propeller which is anodized to retard corrosion. The propeller is 76 inches in diameter.

KNOW THIS FOR THE CHECKRIDE

Fixed pitch means the tachometer is your power gauge (PHAK 7-5).

Source: POH 7-36, PHAK 7-5, PHAK 5-29

Quick Check

AI.II.E.K1c

1 **What engine does the C172S have, and how much power?**

Answer: Lycoming IO-360-L2A, 180 hp at 2,700 RPM (POH 7-28).

2 **What instrument shows engine power in the C172S?**

Answer: The tachometer (PHAK 7-5).

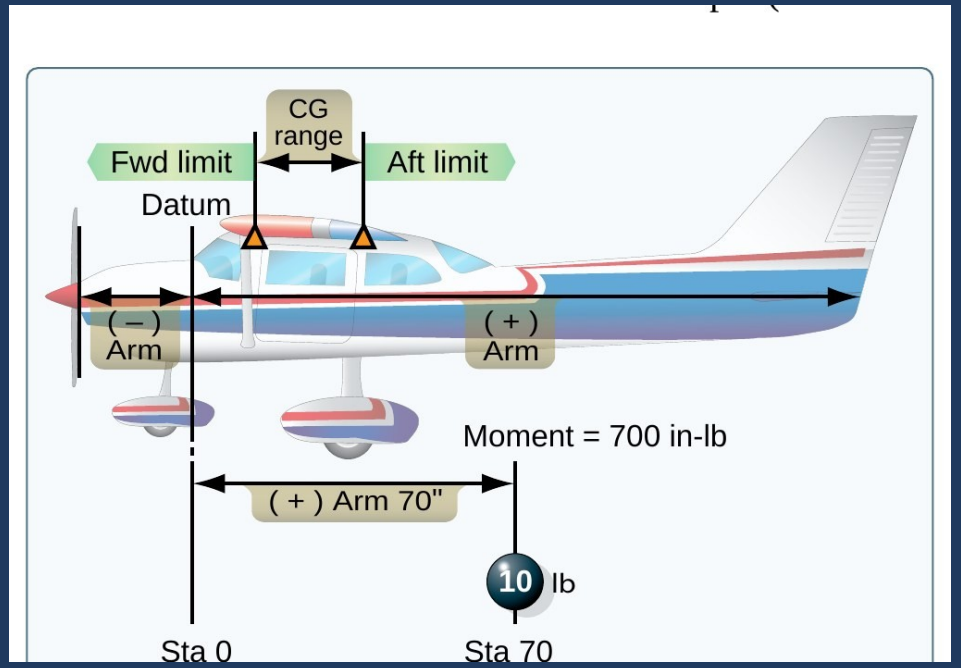
3 **Why is the propeller blade twisted?**

Answer: For even lift from hub to tip, because the tip travels faster (PHAK 7-4).

Source: POH 7-28, PHAK 7-4, PHAK 7-5

AI.II.F.K3

Weight and Balance Terms



What you will learn

- Datum and arm
- Moment and CG
- Basic empty weight and useful load
- CG limits and maximum weight

Why it matters: examiners ask for these word for word. Learn the PHAK wording, then the plain words.

Source: PHAK 10-4, PHAK 10-5

Definitions: Datum and Arm

AI.II.F.K3

DATUM (REFERENCE DATUM)

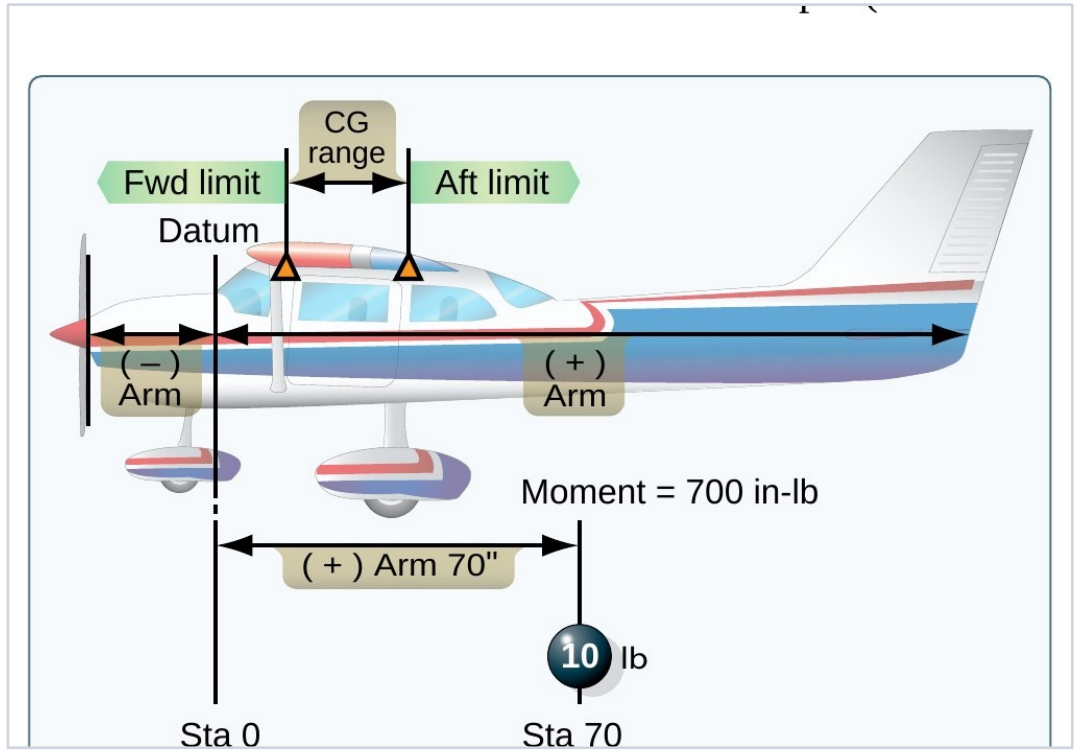
“an imaginary vertical plane or line from which all measurements of arm are taken. The datum is established by the manufacturer. Once the datum has been selected, all moment arms and the location of CG range are measured from this point.”

ARM (MOMENT ARM)

“the horizontal distance in inches from the reference datum line to the CG of an item. The algebraic sign is plus (+) if measured aft of the datum and minus (−) if measured forward of the datum.”

IN PLAIN WORDS

The datum is the zero point on a tape measure. The arm is how far each seat, tank, or bag sits from that zero point.



Source: PHAK 10-4, PHAK 10-5, PHAK Fig 10-2

Definitions: Moment and CG

AI.II.F.K3

MOMENT

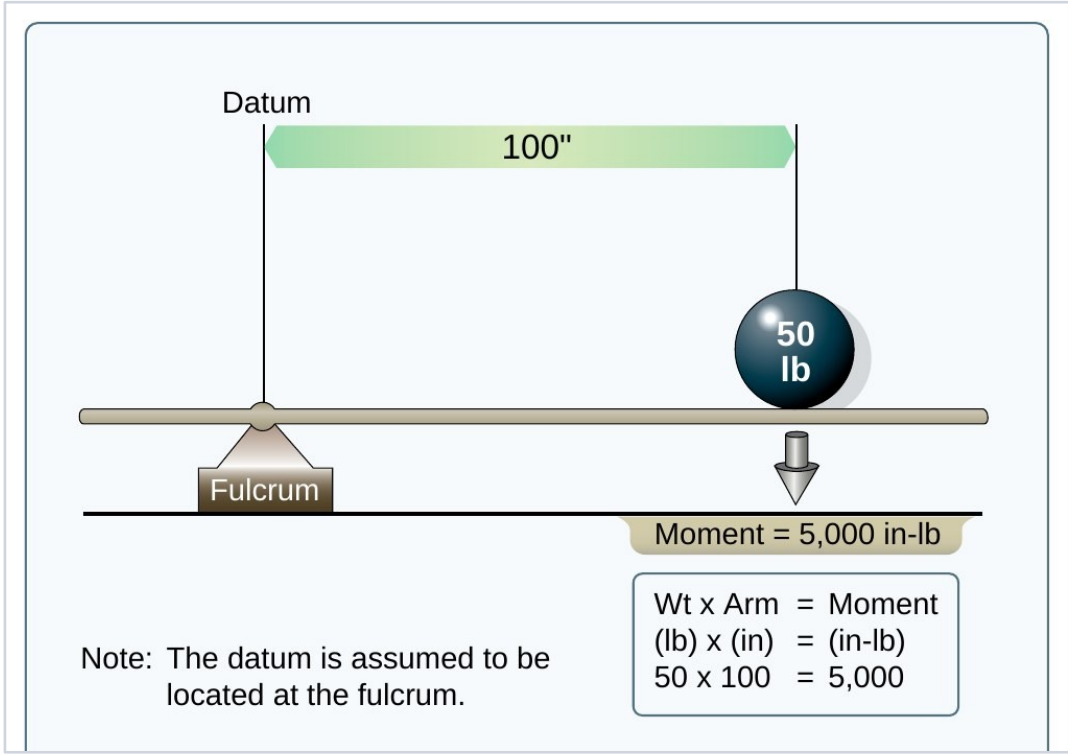
"the product of the weight of an item multiplied by its arm. Moments are expressed in pound-inches (in-lb)."

CENTER OF GRAVITY (CG)

"the point about which an aircraft would balance if it were possible to suspend it at that point."

IN PLAIN WORDS

Moment is how hard a weight tries to tip the airplane. The CG is the balance point of everything combined.



Source: PHAK 10-4, PHAK 10-5, PHAK Fig 10-3

Definitions: Basic Empty Weight and Useful Load

AI.II.F.K3

BASIC EMPTY WEIGHT (GAMA)

"the standard empty weight plus the weight of optional and special equipment that have been installed."

USEFUL LOAD

"the weight of the pilot, copilot, passengers, baggage, usable fuel, and drainable oil. It is the basic empty weight subtracted from the maximum allowable gross weight."

IN PLAIN WORDS

Basic empty weight is the airplane with unusable fuel and full oil, nothing else. Useful load is what is left for people, bags, and fuel.

SAMPLE LOADING PROBLEM

ITEM DESCRIPTION	WEIGHT AND MOMENT TABULATION			
	SAMPLE AIRPLANE		YOUR AIRPLANE	
	Weight (Lbs.)	Moment (Lb-ins. /1000)	Weight (Lbs.)	Moment (Lb-ins. /1000)
1. Basic Empty Weight (Use the data pertaining to your airplane as it is presently equipped. Includes unusable fuel and full oil)	1642	62.6		
2. Usable Fuel (At 6 Lbs./Gal.)				
Standard Fuel 53 Gallons Maximum				
Reduced Fuel (30 Gallons)	180	8.6		
3. Pilot and Front Passenger (FS 32 to 50)	340	12.6		
4. Rear Passengers (FS 74)	340	24.8		
5.* Baggage "A" (FS 82 to 108) 120 Pounds Maximum	56	4.6		
6.* Baggage "B" (FS 108 to 142) 50 Pounds Maximum				
7. **RAMP WEIGHT AND MOMENT**	2558	113.2		
8. Fuel allowance for engine start, taxi and runup	-8.0	-0.4		
9. **TAKEOFF WEIGHT AND MOMENT** (Subtract Step 8 from Step 7)	2550	112.8		

10. Locate this point (2550 at 112.8) on the Center-of-Gravity Moment Envelope, and since this point falls within the envelope, the loading is acceptable.

* The maximum allowable combined weight capacity for baggage in areas A and B is 120 pounds.

Source: PHAK 10-4, PHAK 10-5, POH Fig 6-3

Definitions: CG Limits and Maximum Weight

AI.II.F.K3

CG LIMITS

“the specified forward and aft points within which the CG must be located during flight. These limits are indicated on pertinent aircraft specifications.”

MAXIMUM WEIGHT

“the maximum authorized weight of the aircraft and all of its equipment as specified in the TCDS for the aircraft.”

IN PLAIN WORDS

CG limits are the front and back edges of the safe balance zone. Maximum weight is the heaviest the airplane may be.



CENTER-OF-GRAVITY LIMITS

NORMAL CATEGORY

Center-of-Gravity Range:

Forward:	35.0 inches aft of datum at 1950 pounds or less, with straight line variation to 41.0 inches aft of datum at 2550 pounds.
Aft:	47.3 inches aft of datum at all weights.

Reference Datum: Lower portion of front face of firewall.

Source: PHAK 10-5, POH 2-8

C172S Weight and Balance Numbers

AI.II.F.K3

- **Datum:** front face of firewall
- **Max takeoff:** 2,550 lb
- **CG:** 35.0 to 47.3 in
- **CG = moment / weight**

Normal category: maximum takeoff and landing weight 2,550 lb, ramp 2,558 lb (POH 2-8). Forward CG limit 35.0 inches at 1,950 lb or less, straight line to 41.0 inches at 2,550 lb. Aft limit 47.3 inches at all weights (POH 2-8, TCDS 3A12 p30).

Full detail and worked problems are in the II.F Performance and Limitations lesson.

C.G. Range	Normal Category	
	(1) Aft Limits	47.3 inches aft of datum at 2,550 pounds or less.
	(2) Forward Limits	Linear variation from 41.0 inches aft of datum at 2,550 pounds to 35.0 inches aft of datum at 1,950 pounds. 35.0 inches aft of datum at 1,950 pounds or less.
	Utility Category	
	(1) Aft Limits	40.5 inches aft of datum at 2,200 pounds or less.
	(2) Forward Limits	Linear variation from 37.5 inches aft of datum at 2,200 pounds to 35.0 inches aft of datum at 1,950 pounds. 35.0 inches aft of datum at 1,950 pounds or less.
Empty Wt. C.G. Range	None	
Reference Datum	Lower portion of front face of firewall	
MAC	58.8 inches; Leading edge of MAC 25.9 inches aft of datum	
Leveling Means	Left side of Tailcone at 108.0 inches and 142.0 inches aft of datum	

MEMORY AID

Moment = Weight x Arm. CG = Total Moment / Total Weight.

Source: TCDS 3A12 p30, POH 2-8, PHAK 10-7

Quick Check

AI.II.F.K3

1 **Where is the C172S datum?**

Answer: Lower portion of the front face of the firewall (TCDS 3A12 p30).

2 **A 120 lb bag is 95 inches aft of the datum. What is its moment?**

Answer: 120 x 95 = 11,400 in-lb.

3 **Basic empty weight is 1,642 lb and maximum weight is 2,550 lb. What is the useful load?**

Answer: 908 lb.

Source: TCDS 3A12 p30, PHAK 10-6, POH 6-10

AI.II.H.K1

Ground-Based Navigation

What you will learn

- Identify a VOR
- Read the CDI and HSI
- Track a radial in wind
- VOR checks, NDB, interference



Why it matters: when GPS fails, the VOR is your backup. Your students need to use it without reverse sensing mistakes.

Source: AIM 1-1-2 to 1-1-4, PHAK 16-23 to 16-26

VOR: Identification and Accuracy

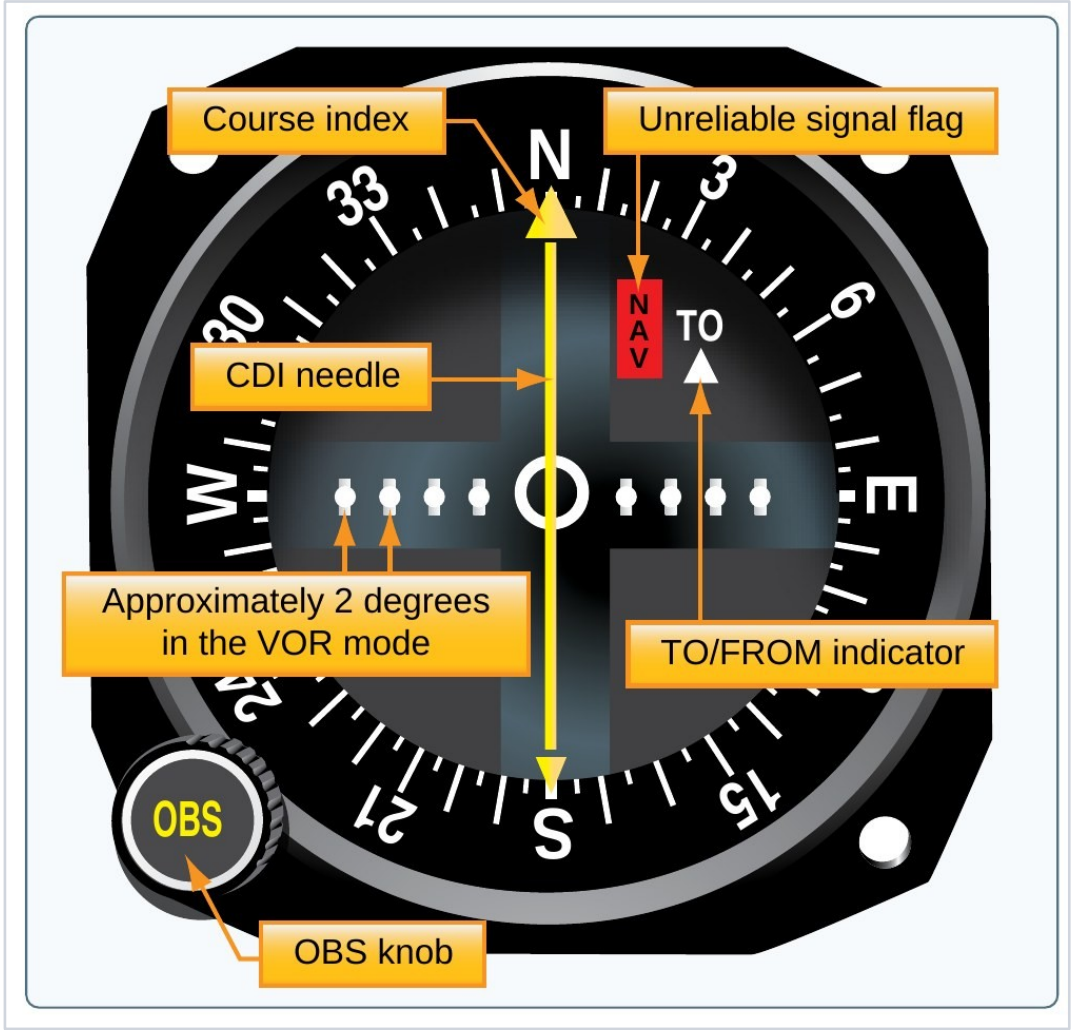
AI.II.H.K1

- **108.0 to 117.95 MHz**
- **Line of sight only**
- **Identify by Morse code**
- **Accuracy:** about 1 degree

VORs operate from 108.0 to 117.95 MHz and are subject to line-of-sight limits. Range increases with altitude (AIM 1-1-3).

Identification: the only positive method is the Morse code identifier or the recorded voice identification with the word "VOR" after the name. During maintenance the station may send T-E-S-T or remove the identifier (AIM 1-1-3).

Accuracy: course alignment is generally plus or minus 1 degree (AIM 1-1-3).



Source: AIM 1-1-3, PHAK Fig 16-29

Orientation: CDI and HSI

AI.II.H.K1

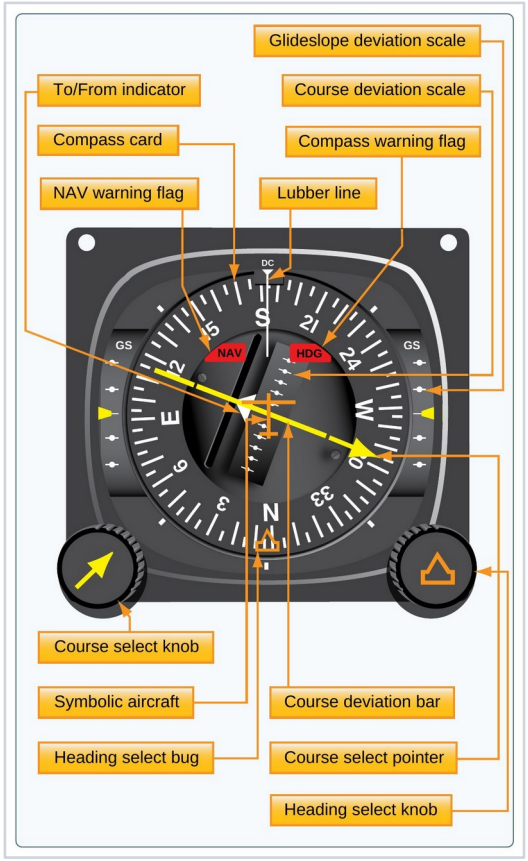
- **OBS sets the course**
- **Needle shows course deviation**
- **TO/FROM shows station side**
- **G1000 shows an HSI**

The CDI has an omnibearing selector (OBS), a course deviation needle, and a TO/FROM indicator (PHAK 16-23).

The HSI combines the heading indicator and the CDI in one display, which is how the G1000 PFD presents it (PHAK 16-24).

Reverse sensing: fly the selected course with a TO indication when going to a station and FROM when going away. If not, the needle works backward (PHAK 16-26).

Study guide fix: the guide says to correct reverse sensing by turning away from the needle. The PHAK way is to set the course so the TO/FROM matches your direction, and then the needle reads normally.





KNOW THIS FOR THE CHECKRIDE

Fly TO when going to the station and FROM when going away, or the needle reverses (PHAK 16-26).

Source: PHAK 16-23, PHAK 16-24, PHAK 16-26, PHAK Fig 16-30

Course Determination: Tracking

AI.II.H.K1

- **Needle shows the course**
- **Turn toward needle, intercept**
- **Hold a wind correction**
- **Wait on small wiggles**

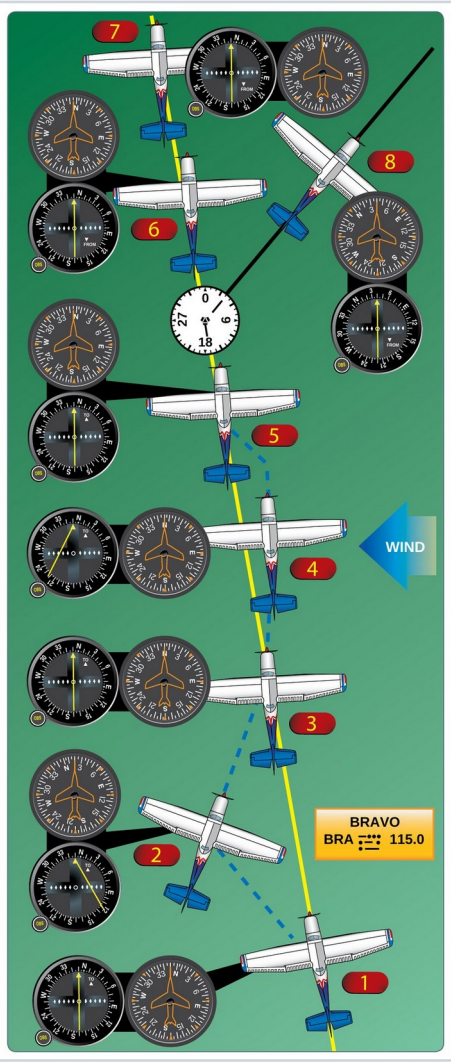
PHAK example: holding 350 degrees with wind from the right, the airplane drifts left and the needle moves right. Turn right to return, then turn back toward the course with a wind correction (PHAK 16-25).

Just turning toward the needle causes overshooting and S-turns. Fly headings that re-intercept, then apply a drift correction. If the needle fluctuates slightly, wait before changing heading (PHAK 16-26).

Station passage: the needle fluctuates, then the TO changes to FROM (PHAK 16-25).

REAL WORLD SCENARIO

Your student keeps chasing the needle and flies S-turns down the radial. Show them how to bracket and hold a wind correction.



Source: PHAK 16-25, PHAK 16-26, PHAK Fig 16-32

Tests and Regulations

AI.II.H.K1

- **VOT:** 0 FROM or 180 TO
- **Ground check:** 4 degrees
- **Airborne check:** 6 degrees
- **IFR:** checked within 30 days

14 CFR 91.171 requires a VOR check within the preceding 30 days for IFR. Limits: VOT or ground checkpoint plus or minus 4 degrees, airborne checkpoint plus or minus 6 degrees, dual VOR check 4 degrees between the two (14 CFR 91.171).

Using a VOT: with the CDI centered, the OBS should read 0 degrees FROM or 180 degrees TO (AIM 1-1-4).

Locations of VOTs and checkpoints are in the Chart Supplement (AIM 1-1-4).

KNOW THIS FOR THE CHECKRIDE

Ground check: 4 degrees. Airborne check: 6 degrees. Dual VOR: 4 degrees. Within 30 days for IFR (14 CFR 91.171).



Source: 14 CFR 91.171, AIM 1-1-4

NDB and ADF

AI.II.H.K1

- **190 to 535 kHz**
- **ADF needle points to station**
- **No warning flag**
- **Keep listening to the ID**

NDBs normally operate from 190 to 535 kHz and send a continuous three-letter Morse identifier, except compass locators (AIM 1-1-2).

ADF receivers do not have a flag to warn of bad bearing information, so the pilot should continuously monitor the NDB identifier (AIM 1-1-2).

The RMI shows ADF and VOR bearings with pointers on a compass card (PHAK 16-25).

KNOW THIS FOR THE CHECKRIDE

The ADF has no warning flag. Keep listening to the identifier (AIM 1-1-2).



Source: AIM 1-1-2, PHAK 16-25, PHAK Fig 16-31

Interference and Signal Integrity

AI.II.H.K1

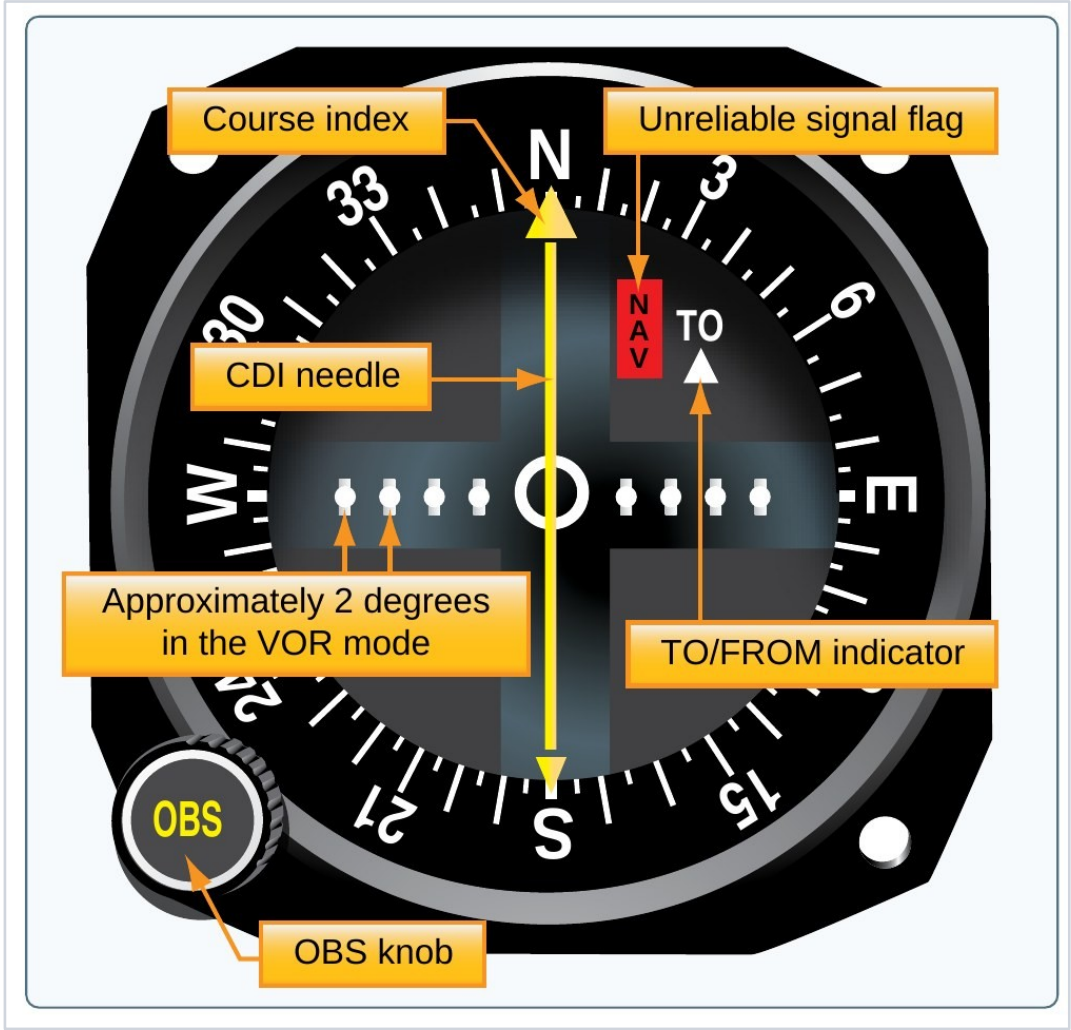
- **Prop RPM can swing CDI**
- **Change RPM slightly**
- **Flag means unreliable signal**
- **NDB:** lightning, night static

Certain propeller rpm settings can make the CDI fluctuate as much as plus or minus 6 degrees. Slight rpm changes normally smooth it out. Check this before reporting a station or receiver problem (AIM 1-1-3).

Minor course roughness or brief flag activity can occur, especially in mountains. Use the TO/FROM indicator to confirm station passage (AIM 1-1-3).

NDB bearings can be disturbed by lightning, precipitation static, and distant stations at night (AIM 1-1-2).

Appropriate use of navigation data: when equipment decodes and displays the identifier, compare it to the chart (AIM 1-1-3).



Source: AIM 1-1-2, AIM 1-1-3

Quick Check

AI.II.H.K1

1 **What is the only positive way to identify a VOR?**

Answer: Its Morse code identifier or recorded voice ID with the word "VOR" (AIM 1-1-3).

2 **With the CDI centered on a VOT, what should the OBS read?**

Answer: 0 FROM or 180 TO (AIM 1-1-4).

3 **The CDI swings back and forth at one RPM. What do you try first?**

Answer: Change the RPM slightly (AIM 1-1-3).

4 **Why must you keep listening to an NDB identifier?**

Answer: The ADF has no warning flag for bad bearings (AIM 1-1-2).

Source: AIM 1-1-3, AIM 1-1-4, AIM 1-1-2

AI.II.I.K3

Plotting a Course

What you will learn

- True course
- Using the plotter
- Magnetic variation
- True to magnetic



Why it matters: every nav log starts with a course line. Get the variation backward and you fly the wrong way.

Source: PHAK 16-5 to 16-7, PHAK 16-12, PHAK 16-18

Definition: True Course

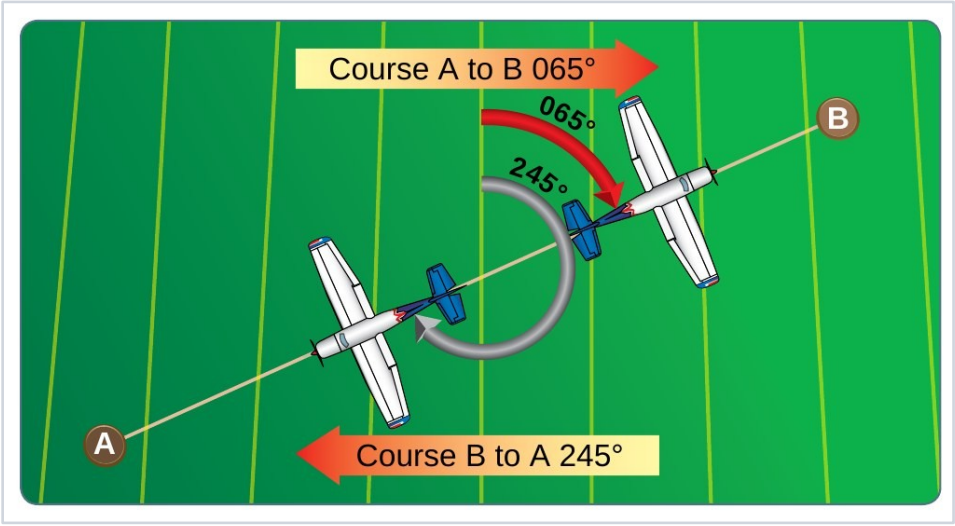
AI.II.I.K3

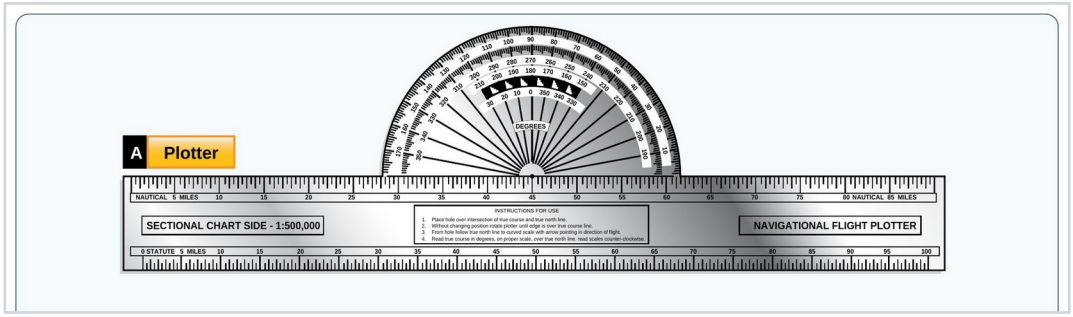
TRUE COURSE (TC)

"The course measured on the chart is known as the true course (TC). This is the direction measured by reference to a meridian or true north (TN)."

IN PLAIN WORDS

The line you draw on the chart, measured from true north with the plotter.





Source: PHAK 16-5, PHAK Fig 16-7

Plotting the Course

AI.II.I.K3

- **Line:** airport center to center
- **Measure at midpoint meridian**
- **Plotter gives true course**
- **Measure distance in NM**

Draw the course line from the center of the departure airport to the center of the destination (PHAK 16-18).

Because meridians converge toward the poles, measure the course at a meridian near the midpoint of the course (PHAK 16-5).

A plotter is a protractor and ruler. Most have NM and SM scales, with a sectional scale on one side and a world aeronautical chart scale on the other (PHAK 16-12). Use the sectional scale on a sectional chart.

Use nautical miles so distance matches knots for time and fuel. The PHAK fuel example: 400 NM at 100 knots GS = 4 hours, at 5 GPH = 20 gallons (PHAK 16-12).

KNOW THIS FOR THE CHECKRIDE

Knots go with nautical miles. Use the NM scale on the plotter.



Source: PHAK 16-5, PHAK 16-12, PHAK 16-18, PHAK Fig 16-25

Definition: Variation

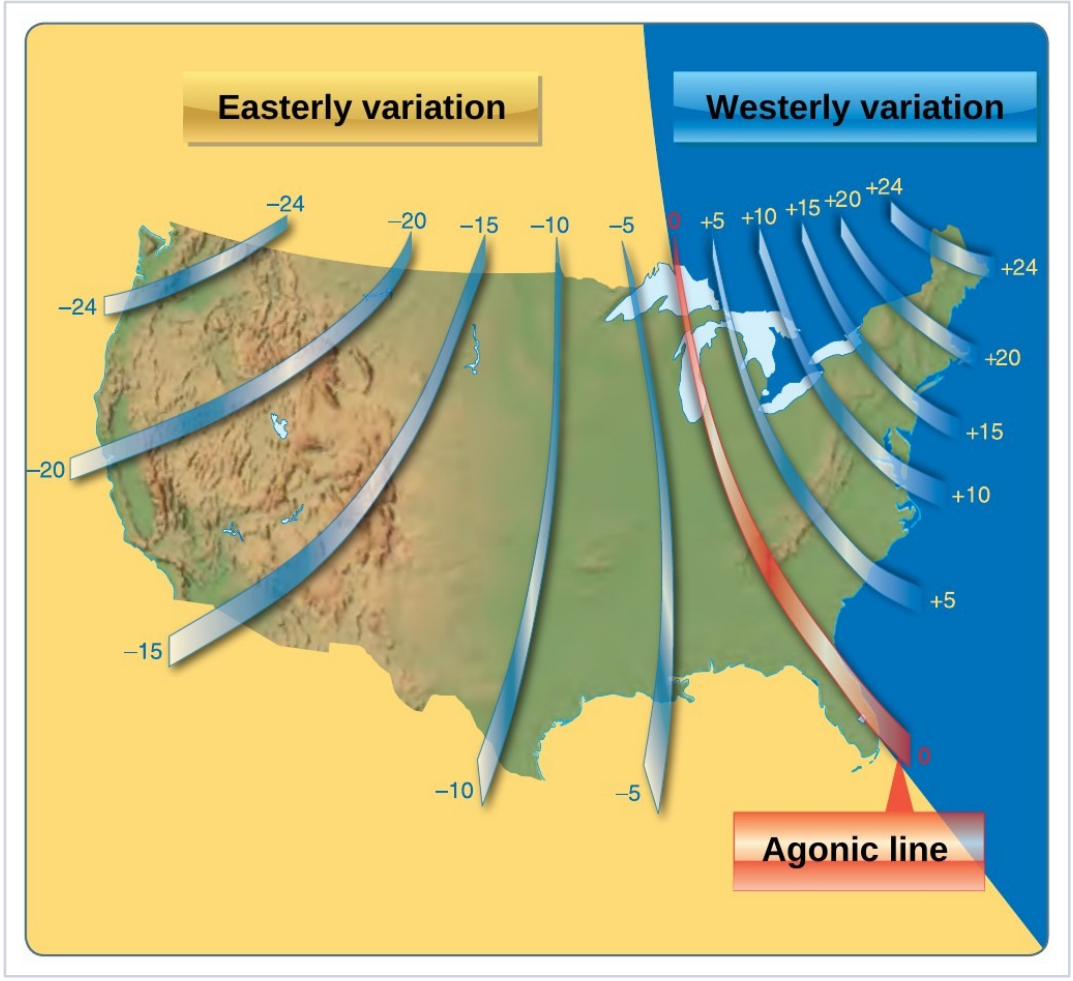
AI.II.I.K3

VARIATION

“Variation is the angle between TN and magnetic north (MN). It is expressed as east variation or west variation depending upon whether MN is to the east or west of TN.”

IN PLAIN WORDS

True north and magnetic north are in different places. Variation is the angle between them, and the chart shows how much.



Source: PHAK 16-6, PHAK Fig 16-9

True Course to Magnetic Course

AI.II.I.K3

- **Use nearest isogonic line**
- **West variation:** add
- **East variation:** subtract
- **East is least, west is best**

Convert TC to magnetic course by adding or subtracting the variation from the nearest isogonic line. If variation is west, add. If east, subtract (PHAK 16-7).

PHAK example: variation 9 degrees E. True north 360 becomes magnetic 351. True east 090 becomes magnetic 081 (PHAK 16-7).

If you cross isogonic lines, use the new value from that point on (PHAK 16-7).

MEMORY AID

East is least (subtract). West is best (add) (PHAK 16-7).



Source: PHAK 16-7

Quick Check

AI.II.I.K3

1 **When you measure with a plotter, are you referencing true or magnetic north?**

Answer: True north. The result is the true course (PHAK 16-5).

2 **True course 090, variation 9 degrees east. Magnetic course?**

Answer: 081 (PHAK 16-7).

3 **On a long route, where do you measure the course?**

Answer: At a meridian near the midpoint (PHAK 16-5).

Source: PHAK 16-5, PHAK 16-7

AI.II.I.K4

Power Setting Selection

What you will learn

- Use the POH cruise tables
- Power in a fixed-pitch airplane
- Read the chart notes
- Speed versus fuel

CRUISE PERFORMANCE TABLE

CONDITIONS:
Standard Conditions
Zero Wind

ALTITUDE	75% POWER		65% POWER		55% POWER	
	KTAS	NMPG	KTAS	NMPG	KTAS	NMPG
Sea Level	114	11.2	108	12.0	101	12.8
4000 feet	119	11.7	112	12.4	104	13.2
8000 feet	124	12.2	117	12.9	107	13.6

Why it matters: the power setting you pick sets your speed, fuel burn, and range for the whole trip.

Source: POH 4-33 to 4-35, POH Fig 5-8, PHAK 7-5

Choosing a Power Setting

AI.II.I.K4

- **Start with POH cruise table**
- **Pick altitude with best winds**
- **Low power saves fuel**
- **High power:** faster, more fuel

The Cruise Performance Table shows true airspeed and nautical miles per gallon at various altitudes and percent powers, for standard conditions and zero wind. Use it with winds aloft to find the most favorable altitude and power setting (POH 4-33).

The POH says choosing altitude for the best winds and using low power settings are significant factors to reduce fuel use on every trip (POH 4-33).

Example from Figure 4-3: at 8,000 feet, 75 percent gives 124 KTAS at 12.2 NMPG and 55 percent gives 107 KTAS at 13.6 NMPG (POH 4-33).

Fuel burn also depends on engine condition, rpm, mixture, and percent of horsepower used (PHAK 16-12).

CRUISE PERFORMANCE TABLE

CONDITIONS:
Standard Conditions
Zero Wind

ALTITUDE	75% POWER		65% POWER		55% POWER	
	KTAS	NMPG	KTAS	NMPG	KTAS	NMPG
Sea Level	114	11.2	108	12.0	101	12.8
4000 feet	119	11.7	112	12.4	104	13.2
8000 feet	124	12.2	117	12.9	107	13.6

KNOW THIS FOR THE CHECKRIDE

Pick the altitude with the best winds, then pick the power. Lower power saves fuel (POH 4-33).

Source: POH 4-33, PHAK 16-12

Power in a Fixed-Pitch Airplane

AI.II.I.K4

- **Tachometer shows engine power**
- **Same RPM:** less power up high
- **Percent of max continuous power**
- **Above 75%:** full rich

In a fixed-pitch propeller airplane, the tachometer is the indicator of engine power (PHAK 7-5).

Power depends on air density. 2,300 rpm at 5,000 feet produces less horsepower than 2,300 rpm at sea level (PHAK 7-5).

The POH cruise data is based on the recommended lean mixture at 75 percent Maximum Continuous Power (MCP) and less (POH 4-34). Use full rich mixture above 75 percent power (POH 4-35).

Study guide fix: the guide explains brake horsepower as power measured with a brake. That explanation is not in the FAA handbooks. The PHAK uses brake horsepower as engine power and says propeller efficiency is thrust horsepower divided by brake horsepower (PHAK 5-29).



REAL WORLD SCENARIO

Your student sets 2,400 RPM at 8,000 feet and expects the same speed as 2,400 RPM at sea level. Ask why the airplane is slower and where to find the real numbers.

Source: PHAK 7-5, POH 4-34, POH 4-35

Example: Read the Notes

AI.II.I.K4

- **8,000 ft, +20 C:** 2,600 RPM
- **64% power, 117 KTAS**
- **Fuel flow:** 8.9 GPH
- **Note:** above 75% full rich

POH sample: at 8,000 feet and 20 degrees C above standard, 2,600 RPM gives 64 percent power, 117 KTAS, and 8.9 GPH (POH 5-5).

Figure 5-8 note: maximum cruise power using recommended lean mixture is 75 percent MCP. Values above 75 percent MCP are shown for interpolation only. Operations above 75 percent MCP must use full rich mixture (POH Fig 5-8).

Teach the learner to read the whole chart, including conditions and notes, before choosing a setting.

KNOW THIS FOR THE CHECKRIDE

Always read the chart notes before you choose a power setting.

CRUISE PERFORMANCE

CONDITIONS:
2550 Pounds
Recommended Lean Mixture At All Altitudes (Refer to Section 4, Cruise)

PRESS ALT FT	RPM	20°C BELOW STANDARD TEMP			STANDARD TEMPERATURE			20°C ABOVE STANDARD TEMP		
		% MCP	KTAS	GPH	% MCP	KTAS	GPH	% MCP	KTAS	GPH
2000	2550	83	117	11.1	77	118	10.5	72	117	9.9
	2500	78	115	10.6	73	115	9.9	68	115	9.4
	2400	69	111	9.6	64	110	9.0	60	109	8.5
	2300	61	105	8.6	57	104	8.1	53	102	7.7
	2200	53	99	7.7	50	97	7.3	47	95	6.9
	2100	47	92	6.9	44	90	6.6	42	89	6.3
4000	2600	83	120	11.1	77	120	10.4	72	119	9.8
	2550	79	118	10.6	73	117	9.9	68	117	9.4
	2500	74	115	10.1	69	115	9.5	64	114	8.9
	2400	65	110	9.1	61	109	8.5	57	107	8.1
	2300	58	104	8.2	54	102	7.7	51	101	7.3
	2200	51	98	7.4	48	96	7.0	45	94	6.7
	2100	45	91	6.6	42	89	6.4	40	87	6.1
6000	2650	83	122	11.1	77	122	10.4	72	121	9.8
	2600	78	120	10.6	73	119	9.9	68	118	9.4
	2500	70	115	9.6	65	114	9.0	60	112	8.5
	2400	62	109	8.6	57	108	8.2	54	106	7.7
	2300	54	103	7.8	51	101	7.4	48	99	7.0
	2200	48	96	7.1	45	94	6.7	43	92	6.4

Source: POH 5-5, POH Fig 5-8

Quick Check

AI.II.I.K4

1. **What should your student look at first to pick a power setting?**
   Answer: The POH cruise performance table and charts (POH 4-33).

2. **Above 75 percent power, how do you set the mixture?**
   Answer: Full rich (POH Fig 5-8).

3. **Same RPM, higher altitude: more or less power?**
   Answer: Less, because the air is less dense (PHAK 7-5).

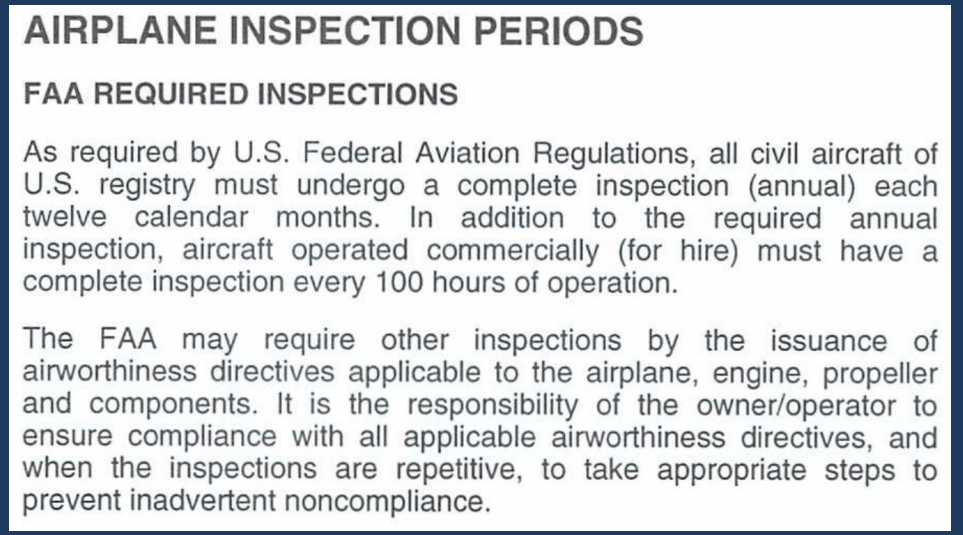
Source: POH 4-33, POH Fig 5-8, PHAK 7-5

AI.III.B.K1b

Inspections and Logbooks

What you will learn

- Annual and 100-hour
- Transponder, ELT, other checks
- ADs
- What goes in the logbook



AIRPLANE INSPECTION PERIODS

FAA REQUIRED INSPECTIONS

As required by U.S. Federal Aviation Regulations, all civil aircraft of U.S. registry must undergo a complete inspection (annual) each twelve calendar months. In addition to the required annual inspection, aircraft operated commercially (for hire) must have a complete inspection every 100 hours of operation.

The FAA may require other inspections by the issuance of airworthiness directives applicable to the airplane, engine, propeller and components. It is the responsibility of the owner/operator to ensure compliance with all applicable airworthiness directives, and when the inspections are repetitive, to take appropriate steps to prevent inadvertent noncompliance.

Why it matters: as a CFI you must confirm the airplane is legal before every lesson. The logbooks are how you prove it.

Source: 14 CFR 91.409, 91.413, 91.417

Required Inspections

AI.III.B.K1b

- **Annual:** 12 calendar months
- **100-hour:** for hire, instruction
- **May exceed 10 hours to shop**
- **Comply with all ADs**

Annual: no person may operate an aircraft unless it has had an annual inspection within the preceding 12 calendar months (14 CFR 91.409). Twelve calendar months runs from any day of a month to the last day of the same month the next year (PHAK 9-8).

100-hour: required to carry any person (other than crew) for hire, or to give flight instruction for hire in an aircraft that person provides. It may be exceeded by not more than 10 hours to reach a place where it can be done, and the extra time counts toward the next 100 hours (14 CFR 91.409).

The owner or operator is primarily responsible for keeping the aircraft airworthy, including compliance with ADs under part 39 (14 CFR 91.403). The POH says the FAA may require other inspections through ADs (POH 8-7).

Note: an amendment to 91.403 and 91.409 was published November 10, 2026 and is not yet in the current text. Recheck before your checkride.

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KNOW THIS FOR THE CHECKRIDE

Annual: every 12 calendar months. 100-hour: carrying people for hire, or paid instruction in an airplane you provide.

Source: 14 CFR 91.409, 14 CFR 91.403, PHAK 9-8, POH 8-7

Other Required Checks

AI.III.B.K1b

- **Transponder:** 24 calendar months
- **ELT:** 12 calendar months
- **Altimeter, static:** 24 (IFR)
- **VOR check:** 30 days (IFR)

Transponder: tested and inspected within the preceding 24 calendar months (14 CFR 91.413).

ELT: inspected within 12 calendar months after the last inspection (14 CFR 91.207).

Altimeter and static system: tested within 24 months to operate in controlled airspace under IFR (PHAK 9-9).

VOR: checked within the preceding 30 days for IFR (14 CFR 91.171).

AIRPLANE INSPECTION PERIODS

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MEMORY AID

Transponder 24, ELT 12, altimeter and static 24 (IFR), VOR 30 days (IFR).

Source: 14 CFR 91.413, 14 CFR 91.207, 14 CFR 91.171, PHAK 9-9

Logbook Documentation

AI.III.B.K1b

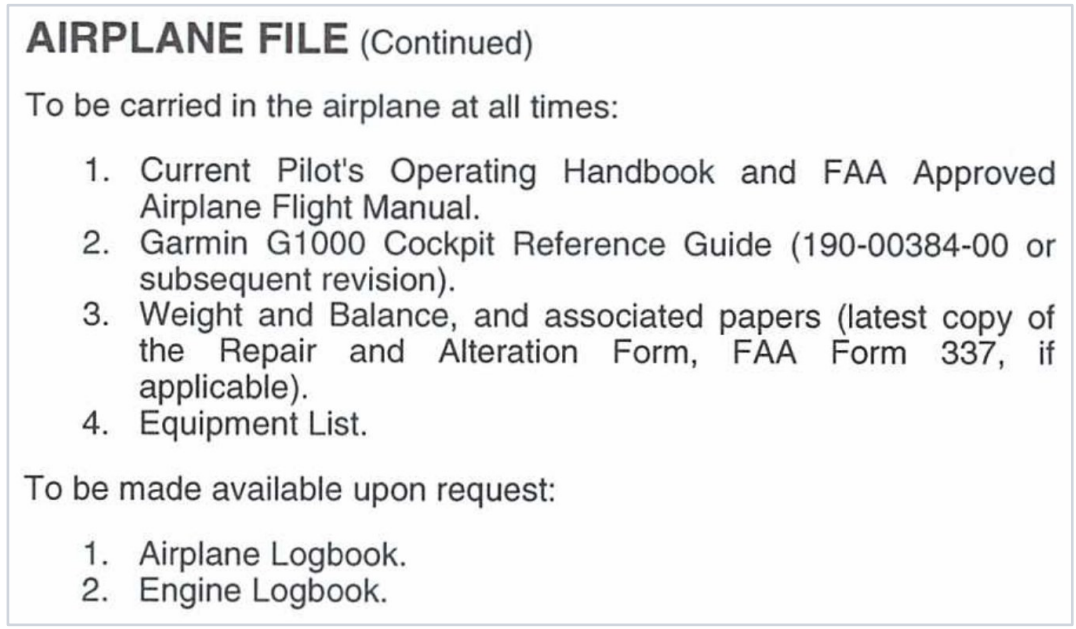
- **Work done, date, signature**
- **Total time, AD status**
- **Kept by owner or operator**
- **Some records stay with plane**

Maintenance records must include a description of the work, the date completed, and the signature and certificate number of the person approving return to service (14 CFR 91.417).

Records must also show total time in service, status of life-limited parts, time since overhaul, current inspection status, and current AD status with method of compliance (14 CFR 91.417).

Work records are kept until the work is repeated or superseded, or for 1 year. Total time, AD status, and similar records are kept and transferred with the aircraft when sold (14 CFR 91.417).

C172S POH: the airplane logbook and engine logbook are to be made available on request (POH 8-7).



AIRPLANE FILE (Continued)

To be carried in the airplane at all times:

1. Current Pilot's Operating Handbook and FAA Approved Airplane Flight Manual.
2. Garmin G1000 Cockpit Reference Guide (190-00384-00 or subsequent revision).
3. Weight and Balance, and associated papers (latest copy of the Repair and Alteration Form, FAA Form 337, if applicable).
4. Equipment List.

To be made available upon request:

1. Airplane Logbook.
2. Engine Logbook.

KNOW THIS FOR THE CHECKRIDE

Every maintenance entry: what was done, the date, and the signature and certificate number.

Source: 14 CFR 91.417, POH 8-7

Quick Check

AI.III.B.K1b

1 **How often is an annual inspection required?**

Answer: Within the preceding 12 calendar months (14 CFR 91.409).

2 **You give paid instruction in an airplane you provide. What other inspection applies?**

Answer: The 100-hour inspection (14 CFR 91.409).

3 **How often must the transponder be checked?**

Answer: Every 24 calendar months (14 CFR 91.413).

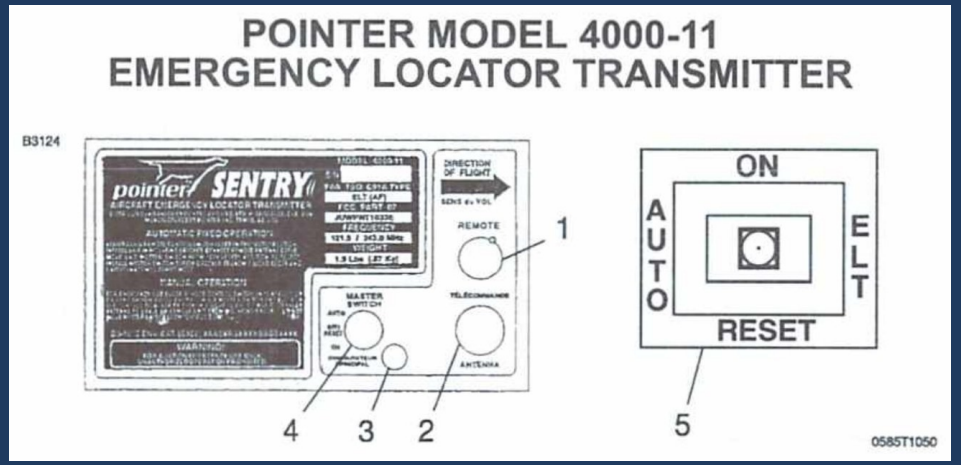
4 **What three things must a maintenance entry include?**

Answer: Description of work, date, and the signature and certificate number of the person approving return to service (14 CFR 91.417).

Source: 14 CFR 91.409, 91.413, 91.417

AI.XII.D.K1

Emergency Locator Transmitter



What you will learn

- How the ELT works
- Using the C172S ELT
- Limitations
- Testing and inspection rules

Why it matters: the ELT is how search and rescue finds you. Your students should know how to turn it on and when they may test it.

Source: AIM 6-2-4, 14 CFR 91.207, POH Supplement 1

ELT Operations

AI.XII.D.K1

- **121.5, 243.0, or 406 MHz**
- **G switch auto activates**
- **Transmits at least 48 hours**
- **406 MHz:** register with NOAA

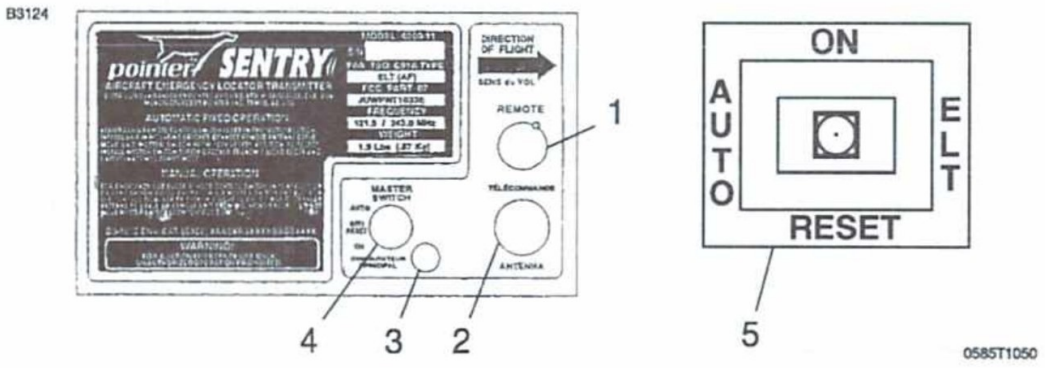
ELTs operate on 121.5 MHz, 243.0 MHz, or 406 MHz. The 406 MHz ELT is digital and can carry owner and aircraft data (AIM 6-2-4).

When armed, ELTs activate automatically from crash forces and transmit continuously for at least 48 hours (AIM 6-2-4).

406 MHz ELTs must be registered with NOAA (AIM 6-2-4).

C172S (this POH): Pointer Model 4000-11. AUTO arms the G switch, ON activates it for test or emergency, OFF/RESET deactivates it (POH S1-7).

POINTER MODEL 4000-11
EMERGENCY LOCATOR TRANSMITTER



CABIN FEATURES

EMERGENCY LOCATOR TRANSMITTER (ELT)

A remote switch/annunciator is installed on the top center location of the right front passenger instrument panel for control of the ELT from the flight crew station. The annunciator, which is in the center of the rocker switch, illuminates when the ELT transmitter is transmitting. The ELT emits an omni-directional signal on the international distress frequencies of 121.5 MHz and 243.0 MHz. General aviation and commercial aircraft, the FAA and CAP monitor 121.5 MHz, and 243.0 MHz is monitored by the military. For a basic overview of the ELT, refer to the Section 9, Supplement 1.

Source: AIM 6-2-4, POH S1-7

Using the C172S ELT

AI.XII.D.K1

- **Remote switch on panel**
- **Before forced landing:** ON
- **Rescue seen:** RESET, AUTO
- **After rescue:** AUTO

The remote switch and annunciator are on the top center of the right front passenger panel. The light comes on when the ELT transmits (POH 7-71).

Before a forced landing, especially in remote or mountain areas, set the remote switch to ON (POH S1-8).

After landing: verify activation and listen on 121.5. Save the battery until you see rescue aircraft. Then set RESET and release to AUTO, and try to call on 121.5. If no contact, go back to ON (POH S1-9).

CABIN FEATURES

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REAL WORLD SCENARIO

After a forced landing in a field, your student is about to call on the radio with the ELT running. Walk them through the POH steps.

Source: POH 7-71, POH S1-8, POH S1-9

ELT Limitations

AI.XII.D.K1

- **Satellites stopped 121.5 in 2009**
- **121.5:** needs ATC or aircraft
- **Must survive the crash**
- **False alarms slow response**

Cospas-Sarsat stopped monitoring 121.5 and 243.0 MHz in 2009. Those ELTs depend on ATC or an overflying aircraft hearing the signal (AIM 6-2-4).

SAR forces wait for confirmation on 121.5/243.0 alerts, which can take hours, but can respond to 406 MHz alerts in minutes (AIM 6-2-4).

An ELT helps only if it survives the crash and activates (AIM 6-2-4).

False alarms cause expensive searches and can hide real signals. Monitor 121.5 before engine shutdown at the end of each flight (AIM 6-2-4).



CABIN FEATURES

EMERGENCY LOCATOR TRANSMITTER (ELT)

A remote switch/annunciator is installed on the top center location of the right front passenger instrument panel for control of the ELT from the flight crew station. The annunciator, which is in the center of the rocker switch, illuminates when the ELT transmitter is transmitting. The ELT emits an omni-directional signal on the international distress frequencies of 121.5 MHz and 243.0 MHz. General aviation and commercial aircraft, the FAA and CAP monitor 121.5 MHz, and 243.0 MHz is monitored by the military. For a basic overview of the ELT, refer to the Section 9, Supplement 1.

KNOW THIS FOR THE CHECKRIDE

Satellites stopped listening to 121.5 MHz in 2009. A 121.5 ELT depends on someone hearing it (AIM 6-2-4).

Source: AIM 6-2-4

ELT Testing and Rules

AI.XII.D.K1

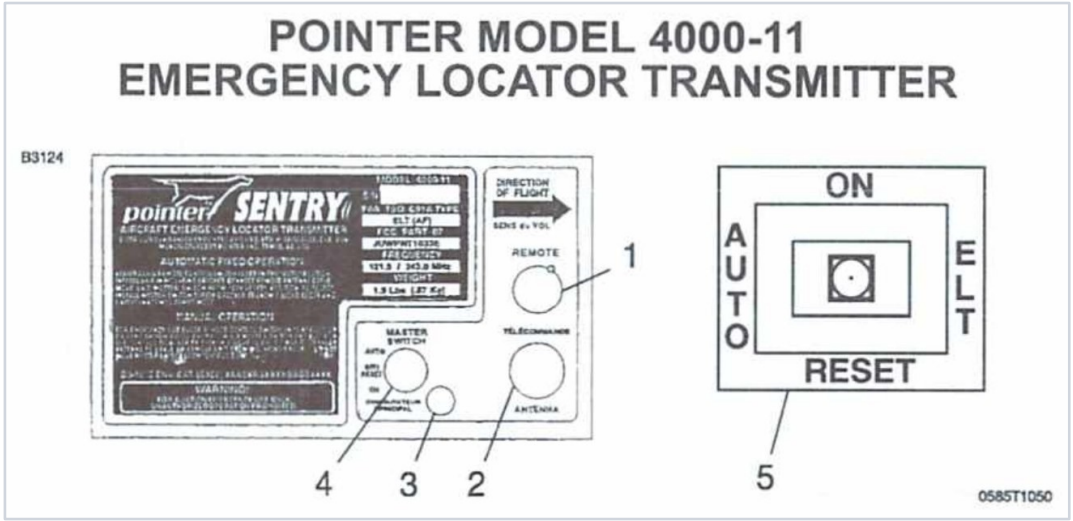
- **Test first 5 minutes of hour**
- **No more than 3 sweeps**
- **Inspect every 12 calendar months**
- **Battery:** 1 hour or 50% life

Analog 121.5/243 ELTs: test only during the first 5 minutes after any hour, for no more than three audible sweeps. Digital 406 MHz ELTs: test per the manufacturer. Airborne tests are not authorized (AIM 6-2-4).

C172S POH: test every 100 hours, within the first 5 minutes of the hour, permitting only three emergency tones (POH S1-10).

14 CFR 91.207: inspect within 12 calendar months for installation, battery corrosion, controls and crash sensor, and signal. Replace or recharge the battery after more than 1 cumulative hour of use or at 50 percent of its useful life. Mark the expiration date on the transmitter and in the maintenance record.

Exceptions include training operations conducted entirely within 50 nautical miles of the airport where they began, and an ELT removed for repair for up to 90 days with a logbook entry and an "ELT not installed" placard (14 CFR 91.207).



MEMORY AID

5, 3, 12, 1, 50: first 5 minutes, 3 sweeps, 12 month inspection, 1 hour of use, 50 percent battery life.

Source: AIM 6-2-4, 14 CFR 91.207, POH S1-10

Quick Check

AI.XII.D.K1

1 **When may you test an analog ELT?**

Answer: In the first 5 minutes after the hour, no more than three sweeps (AIM 6-2-4).

2 **When must the ELT battery be replaced?**

Answer: After more than 1 cumulative hour of use, or at 50 percent of its useful life (14 CFR 91.207).

3 **You are about to make a forced landing in the mountains. What do you do with the ELT?**

Answer: Set the remote switch to ON (POH S1-8).

Source: AIM 6-2-4, 14 CFR 91.207, POH S1-8

Study Guide Fixes

AI.II.E.K1c, AI.II.F.K3, AI.II.H.K1, A...

- **Throttle:** fuel-air flow, not fuel
- **C172S:** no propeller control
- **Fix reverse sensing with OBS**
- **Use PHAK W&B wording**

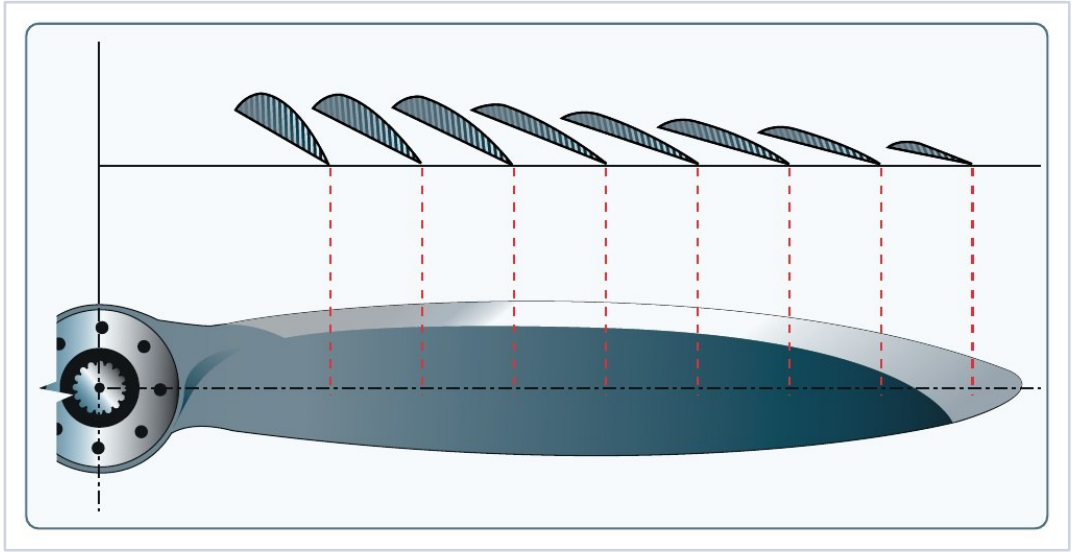
Throttle controls fuel-air flow (PHAK 7-5).

The C172S has a fixed-pitch propeller and no propeller control (POH 7-36).

For reverse sensing, set the OBS so TO/FROM matches your direction of flight (PHAK 16-26).

Use PHAK definitions for weight and balance terms (PHAK 10-4, PHAK 10-5).

Brake horsepower: teach it as engine power. The "measured with a brake" explanation is not in the FAA handbooks (PHAK 5-29).

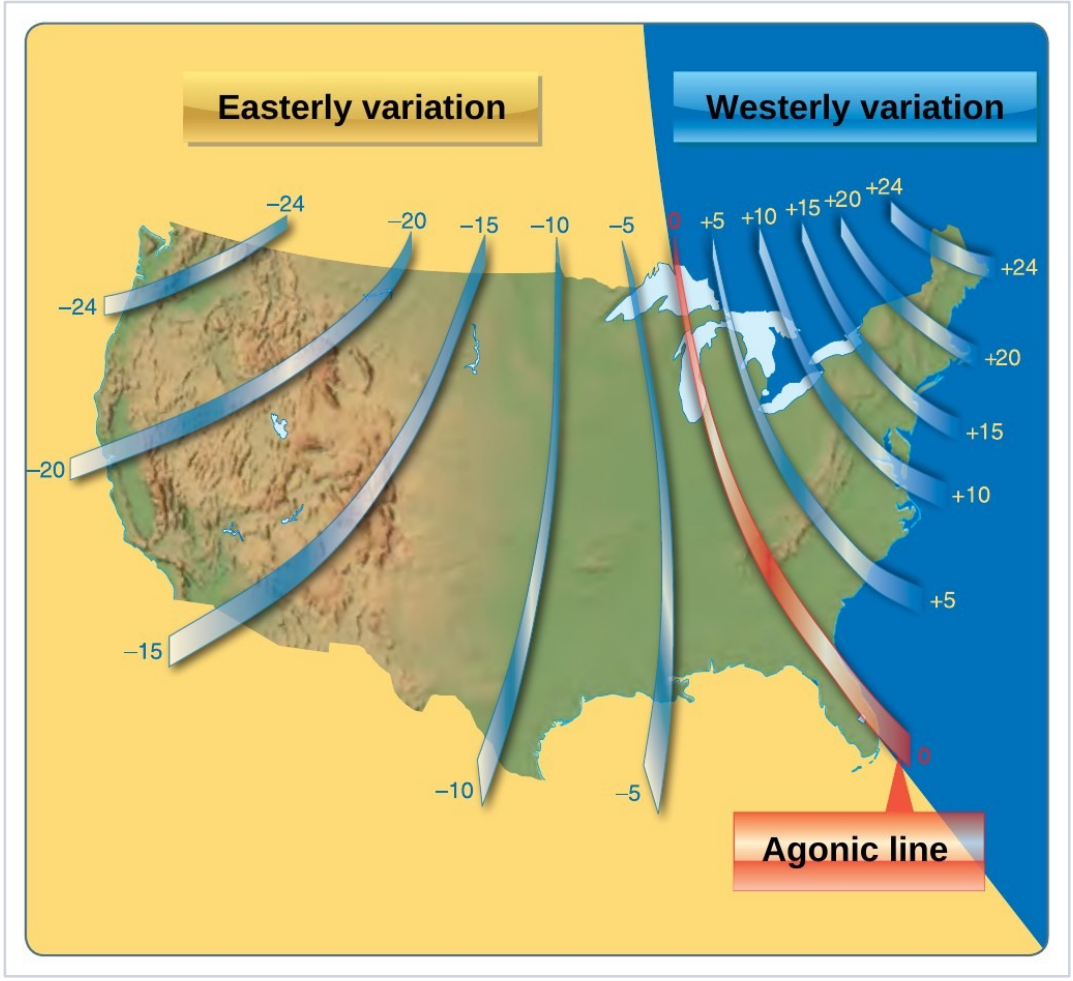


Source: PHAK 7-5, POH 7-36, PHAK 16-26, PHAK 10-4, PHAK 5-29

Review Questions

Review

- **Name the perception factors**
- **Positive vs. neutral static?**
- **East variation:** add or subtract?
- **When may you test an ELT?**

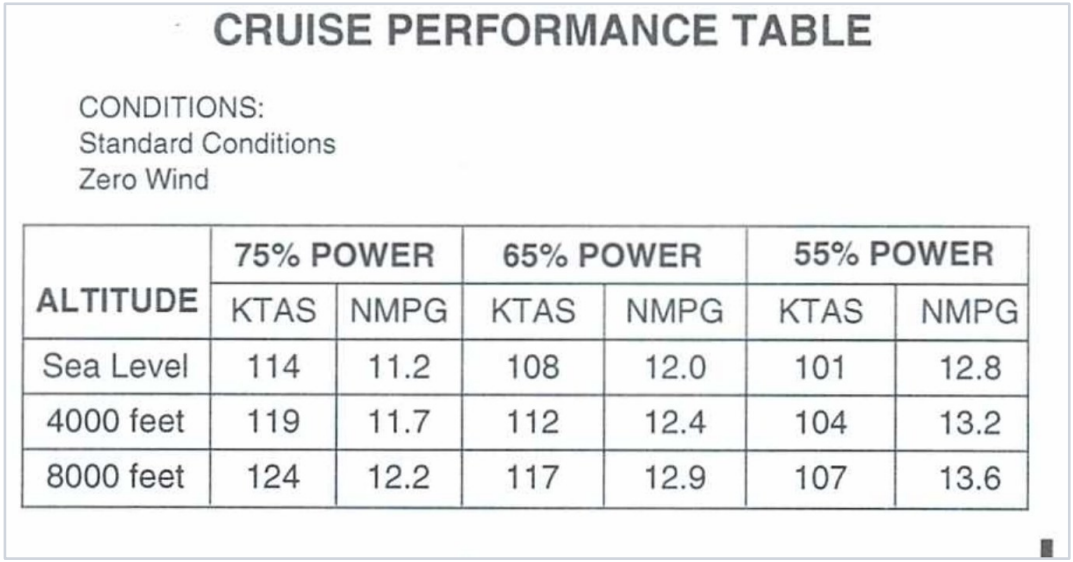


Source: AIH 3-7, PHAK 5-14, PHAK 16-7, AIM 6-2-4

References

Review

- AIH, PHAK, AIM
- 14 CFR Part 91
- C172S POH, TCDS 3A12
- FAA-S-ACS-25



CRUISE PERFORMANCE TABLE

CONDITIONS:
Standard Conditions
Zero Wind

ALTITUDE	75% POWER		65% POWER		55% POWER	
	KTAS	NMPG	KTAS	NMPG	KTAS	NMPG
Sea Level	114	11.2	108	12.0	101	12.8
4000 feet	119	11.7	112	12.4	104	13.2
8000 feet	124	12.2	117	12.9	107	13.6

Source: See lesson plan for full list